

Climate Change and Adaptation in Global Supply-Chain Networks

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This paper examines how physical climate exposure affects firm performance and global supply chains. We document that heat at supplier locations reduces the operating income of suppliers and their customers. Further, customers respond to perceived changes in suppliers' exposure: when suppliers' realized exposure exceeds ex ante expectations, customers are 7% more likely to terminate supplier relationships. Consistent with experience-based learning, this effect increases with signal strength and repetition and decreases with country-level climate adaptation. Subsequent replacement suppliers show a lower expected and realized but similar projected heat exposure. We find similar results for suppliers' exposure to floods. (*JEL* G15, Q54, G30, F64)

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Businesses face increasing pressure to address their operational exposure to physical climate hazards. While managers and investors are looking for ways to alleviate climate change risks by adapting their operations (Lin et al. 2020) and investments (Krueger et al. 2020; Ilhan et al. 2023), academic research has primarily studied how transitory weather shocks affect firms' earnings (Addoum et al. 2020), stock returns (Cuculiza et al. 2023), and labor and capital productivity (Graff-Zivin et al. 2018; Zhang et al. 2018), among others. Much less is known about how firms adapt to gradual change, despite the fact that firms' endogenous responses are crucial to understanding the long-run effects of climate change on financial market outcomes.

Operational risk management in response to climate change is particularly important for firms engaged in extensive international production networks. In a globalized economy, supply chains often move through parts of the world that are most vulnerable to the impact of climate change. As a result, firms might be indirectly exposed to physical risks due to their suppliers. Reflecting these concerns, over 50% of CEOs mentioned risks posed to their global supply chains by climate change as one of their primary concerns in a recent survey (PWC 2015).

However, adapting to climate change is a complex task for economic agents in general and firms in supply-chain organizations in particular. Climate change is characterized by unknowable uncertainty – particularly in the short and medium runs – as weather realizations provide a noisy signal of potential changes in the underlying distribution (Deryugina 2013; Kala 2019). Further, indirect exposure to climate change due to suppliers and customers can be challenging to identify. In this environment, it is unclear how gradually changing exposure to climate hazards could affect firms' decisions to discontinue existing and begin new supply-chain relationships.

In this paper, we study whether firms adjust their supply-chain networks in response to perceived increases in their suppliers' heat exposure.¹ After estimating how the financial consequences of adverse weather propagate from suppliers to their corporate customers around the world, we investigate whether and how firms adapt their supply-chain organizations in response to changes in supplier exposure. In particular, we examine how discrepancies between realized and expected exposure to heat affect the continuation of existing and the initiation of new supply-chain relationships. Our main contribution is to show that customers are more likely to terminate suppliers when their exposure increases beyond historical expectations, and switch to less exposed replacement suppliers. Thereby, the results provide new evidence that climate change could affect firms' operational risk management and the formation of global supply chains.

¹ We do not directly measure customers' priors or expectations. "Perceived increases" refer to short-run increases in realized heat days beyond historical expectations which customers could plausibly observe at supplier locations.

We combine detailed firm-level supply-chain data from FactSet Revere with financial performance data from Worldscope, headquarters and establishment locations from FactSet Fundamentals and Orbis, and data on local temperatures and temperature projections computed in the fifth phase of the Coupled Model Intercomparison Project (CMIP5) from the European Center for Medium-term Weather Forecasts (ECMWF). Our supply-chain data set includes 5,628 (8,200) unique supplier (customer) firms across 92 (74) countries around the world over the period from 2003 to 2016.

We focus on heat as the most pervasive projected gradual trend under climate change. Previous studies have documented several channels through which temperatures affect firm productivity. For example, heat reduces worker performance (Graff-Zivin et al. 2018), labor supply (Graff-Zivin and Neidell 2014), and firm-level output (Zhang et al. 2018), with sharp declines at temperatures over 30°C. Further, anecdotal evidence suggests that heat can constrain water supply, distort transportation, and wreak havoc on electrical grids. The resultant risks to firm operations are expected to increase. For our sample, the number of days on which temperatures exceed 30° C are projected to rise from currently 2.7% of days to more than 12% of days by 2100 without substantial efforts to reduce emissions according to CMIP5 model output.

Before examining firms' supply-chain adaptations, we document how heat exposure affects the financial performance of suppliers and their downstream customers. While Barrot and Sauvagnat (2016) and Carvalho et al. (2021) show that the effect of natural disasters propagates through production networks, it is unclear if moderate realizations of adverse weather, which are projected to gradually increase in frequency and severity, have similar effects. We construct location-specific measures of suppliers' heat exposure based on daily temperatures over a quarter or year at their headquarters. In robustness tests, we measure heat across all supplier locations. Consistent with Somanathan et al. (2021), Zhang et al. (2018), and Pankratz et al. (2023), we find a significant negative effect of heat on supplier operating performance. Further, we document repercussions on downstream customers: following the occurrence of high temperatures in a given firm-quarter at supplier locations, customer operating income over assets decreases by 0.6% relative to the mean. Whereas these effects appear moderate relative to large-scale natural disasters, they could be large enough as financial incentives for customers to consider adapting their supply chains to gradual change.²

Our main analysis focuses on the question of whether firms attempt to learn about climate change and adapt to gradual increases in heat exposure. Given that short- and medium-term weather realizations provide noisy signals for the underlying distribution, detecting change is challenging. However, prior

² We implement a series of robustness tests using alternative heat measures and explore the cross-sectional heterogeneity, for example, with respect to geographic concentration, climate vulnerability, adjustment costs, and supply-chain diversification. We also estimate the tests for suppliers' exposure to floods.

research in finance and economics has proposed experience-based Bayesian updating to model learning in general (Alevy et al. 2007; Chiang et al. 2011), and about climate change in particular (Kelly et al. 2005; Deryugina 2013; Moore 2017; Kala 2019; Choi et al. 2020). In this paper, we test if observed terminations of supply-chain relationships are consistent with an attempt of corporate customers to learn about climate change based on short-run increases in the heat exposure of their suppliers.

We assume that customer firm managers carefully consider and trade off the expected costs and benefits, such as the exposure to environmental hazards, product quality, and input prices of prospective suppliers, based on observable characteristics when entering a supply-chain relations. Under this setting, customer firms have no incentive to alter their supply-chain relations in response to the occurrence of heat events as long as the realizations are in line with ex ante expectations. However, the underlying distribution of adverse weather is projected to change in heterogeneous ways across geographies due to climate change. Hence, if customers perceive an increase in the likelihood of occurrences at their supplier firm, a previously optimal supplier may no longer be optimal going forward, and existing supplier relations may be terminated more frequently.

To test this idea, we construct a new measure of *realized versus expected exposure* by comparing heat days before and during a given supply-chain relationship. Our results show that a supply-chain link is 7.4% more likely to be terminated in a given year if the realized exposure to heat exceeds proxies of customers' ex-ante expectations. This result is robust to alternative periods over which customers form priors, and holds after controlling for industry and country-by-time fixed effects for suppliers and customers. Consistent with experience-based learning about climate change (e.g., Deryugina 2013), we find stronger results after the first relationship-year in which deviations from ex ante expectations would be particularly challenging to interpret. We also document that the likelihood of terminations continues to increase with the repeated exceedance of likely ex ante expectations and when deviations of priors and realizations are large.³

To explore climate change adaptation as the mechanism underlying these results, we use cross-sectional variation in adaptation readiness across supplier countries. Based on data from the Notre-Dame Global Adaptation Index (ND-GAIN) on the economic, governance, and social capacity to adapt to climate change, we find that the likelihood of supplier-termination is higher when adaptation readiness in the supplier country is low. This finding is consistent with the idea that suppliers in low adaptation capacity countries pose a higher future exposure to customers.

³ As we cannot directly observe firms' perceptions, we infer priors from historical expectations at the start of supply-chain relationships and assume that customers observe realizations at the locations of suppliers.

In line with the notion that customers actively adapt to perceived change in climate exposure, we document adjustments along other dimensions. Customers increase their inventory, cash holdings, and R&D investments when the realized exposure to heat exceeds *ex ante* expectations. We find weak evidence pointing toward an additional effect on supplier diversification, that is customers increase the number of other suppliers in the industry of a supplier when realizations of heat exceed plausible expectations. In contrast, we find no evidence for increases in M&A activity. Further, the effect is stronger in the cross-section for suppliers in competitive industries and weaker for closely integrated supply chains and high customer reliance on supplier inputs. The results remain similar when we implement our tests as linear probability models, logistic regressions, and as Cox proportional hazard models.

A potential concern with our interpretation of the results is that supplier terminations may be solely driven by physical supplier-disruptions without customers' attributing those disruptions at least partly to suppliers' increased climate risk exposure. Arguably, it is impossible to prove if and how exactly customers update their beliefs after witnessing supply-chain disruptions. However, and consistent with the notion that customers terminate relationships (only) when they believe that suppliers' climate exposure is persistent, we find that transitory supplier exposures, that is adverse realizations of heat regardless of expectations from a benchmark period, do not lend themselves to subsequent terminations.

While our main tests reflect the idea that firms form priors and update their beliefs based on experienced change, we also consider the role of long-term temperature projections from the CMIP5 project. From the ECMWF, we obtain projections for the average number of heat days between 2040 and 2059. We then estimate our main tests for subsamples of suppliers for which long-term climate models project limited change under various emission trajectories. We find that customers respond to increases in adverse weather above *ex ante* expectations even when long-run projections indicate little to no future change, which may emphasize the challenges of learning about climate change based on experienced change.

Last, we examine whether firms consider suppliers' potential exposure when switching to new suppliers. For this purpose, we identify replacement suppliers as firms with identical SIC codes as the terminated suppliers which begin a supplier-relationship with the same customer within one year. We estimate linear probability models to test whether replacement suppliers have a lower exposure to heat than terminated suppliers conditional on an increased exposure during the terminated supplier relationship. We find a positive effect of climate exceedance on the likelihood that customers choose a replacement supplier with lower *ex post* exposure observed both *during* and *after* the initial relationship. An unexpectedly high exposure during the initial relationship increases the probability that customers choose a less exposed replacement by

6 to 10 percentage points, controlling for industry- and country-specific time fixed effects of both suppliers and customers. We find a smaller, less precisely estimated effect when considering differences in long-term projections.

Our paper contributes to the literature on climate change, finance, and economics along several dimensions. First, we provide novel evidence on the implications of climate change for firms and investors. Previous research in finance has studied how weather shocks directly affect firms and investors (e.g., firm profitability (Zhang et al. 2018; Addoum et al. 2020; Pankratz et al. 2023), housing prices (Baldauf et al. 2020), stock returns (Cuculiza et al. 2023), financial markets (Hong et al. 2019; Schlenker and Taylor 2019), and capital structure (Ginglinger and Moreau 2022)). In contrast, our paper studies how firms *respond* to perceived, gradual changes in climate risk in a unique empirical setting, by focusing on firms' adaptation in the context of global supply-chains. Further, our motivating tests show that firms can be *indirectly* exposed to physical climate hazards, such as heat through their global supplier network. This aspect of our findings connects to Barrot and Sauvagnat (2016), Boehm et al. (2019), and Carvalho et al. (2021), who document the propagation of natural disasters along firm linkages in the United States. Relative to these papers, our findings provide evidence that heat exposure, which causes less severe physical damages but may intensify with long-term, gradual climate change, also propagates through production networks.

Second, our paper is among the first in the finance and environmental economics literature to study how firms learn about and adapt to climate change. For example, Lin et al. (2020), Li et al. (2020), and Li et al. (2023) examine the shift of electricity providers to more flexible power plants, changes in local employment composition, and green innovations in response to changes in local climate risk, respectively. Our paper contributes to this emerging literature by studying firms' responses to *indirect* climate change exposure through supply chains, guided by a conceptual framework of firm production and experience-based learning about climate change.

Third, we contribute to the literature on learning about climate change. Choi et al. (2020) find that market participants revise their beliefs about climate change when experiencing unusual weather using Google search data. Deryugina (2013) uses survey data on beliefs about global warming to show that local temperature fluctuations affect these beliefs in a Bayesian framework. Moore (2017) and Kala (2019) develop and test theoretical models on experience-based learning about climate change. Our work builds on these models, but explores learning and adaptation outside of the agricultural sector, combining both observed signals and climate projections.

Our main finding on the adaptation of supply chains has potentially important implications, as the areas of the world which are disproportionately affected by the impact of climate change are already less developed today (Burke et al. 2015; Carleton and Hsiang 2016). Our findings also speak to the growing literature on endogenous production networks and

macroeconomic growth (e.g., [Antràs et al. 2017](#); [Lim 2018](#); [Oberfield 2018](#); [Acemoglu and Azar 2020](#)).

1. Data Sources and Descriptive Statistics

1.1 Global supply chains

We use international data on supply-chain relationships from FactSet Revere. The data are hand-collected, for example from annual reports, SEC filings, investor presentations, websites, press releases, supply contracts, and purchase obligations. Previous research (e.g., [Hertzel et al. 2008](#); [Cohen and Frazzini 2008](#); [Banerjee et al. 2008](#)) has primarily relied on SEC regulation S-K, which requires U.S. firms to disclose customers representing at least 10% of their sales. Because of this cutoff, start and end dates cannot be distinguished from changes in sales around the reporting threshold. Further, the global coverage in FactSet is important for our study, as supply chains often move through parts of the world outside of the United States that are most vulnerable to climate change. In total, we observe 8,200 (5,769) customer (supplier) firms from 74 (92) countries, comprising almost 595,000 pair-year-quarter observations from 2003 to 2016. Figure 1 shows that most suppliers are located in Asia (40%), North America (39%), and Europe (17%) and operate in manufacturing (SIC 2/3) or transport and utilities (SIC 4).⁴

1.2 Accounting data and firm characteristics

We obtain firm financial data from 2000 to 2016 from Worldscope, with quarterly operating income scaled by lagged total assets as our main measure of performance. In addition, we collect data on revenues, employees, asset tangibility, operating margins, inventories, accounts receivables, cost of goods sold (COGS), spending on research and development (R&D), and delisting dates from Worldscope and Datastream. We construct measures of industry competitiveness as the number of firms and the Herfindahl-Hirschman Index (HHI) of revenues in a given SIC two-digit industry. From the Bureau of Economic Analysis (BEA), we obtain global input-output matrices for 2012 to construct industry-level input concentration as value of sales from supplier-to customer industry and the HHI of dollar values across all input industries for each customer industry. For comparability, we convert all outcomes into U.S. dollars, and trim variables above (below) the 99th (1st) percentile to remove outliers. We exclude firms in the financial industry (SIC 6).

Worldscope covers 99% of the global market capitalization. In contrast, our sample is subset of supply chains and limited to listed firms with financial reports. Moreover, international disparities in the availability of financial data may bias the sample toward developed markets. Compared to the universe of

⁴ See Table IA.1a for the number of firms by region and Table IA.1b for the industry composition of the sample.

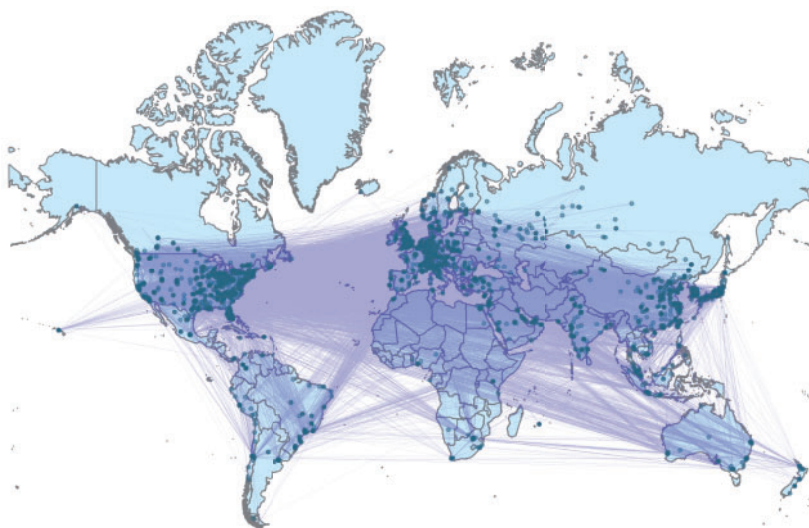


Figure 1
Geography of customer-supplier linkages

This figure shows the connections of the headquarters of the customers and suppliers in our sample. Supply-chain relationships and firm locations are obtained from FactSet Revere, FactSet Fundamentals, and Orbis. [Table IA.1](#) corresponds to the plot and reports the number of customers and suppliers by geographic regions.

firms in *Worldscope*, the observed suppliers are similar to the average firms in their home countries in size (book assets and market capitalization) but have higher revenues over assets. In contrast, customer firms are significantly larger than the average firm in their home country.⁵

We use addresses of headquarters from FactSet as our primary measure of firms' locations. However, plants and establishments might be remote from headquarters. Hence, we collect additional facility-level locations from Orbis. In total, we obtain 1.1 million addresses of incorporated subsidiaries, branches, and establishments. We geocode city, ZIP code, and street names using Bing Maps. As a measure of concentration, we calculate the share of establishments located within a 30-km radius of firms' headquarters. In the main tests, we exclude dispersed firms with fewer than 10% of assets within 30 km of the firms' headquarters. The cutoff follows [Barrot and Sauvagnat \(2016\)](#), who limit their sample to firms with at least 10% of employees at the headquarter

⁵ [Tables IA.1c](#) and [IA.1d](#) show the comparison. For supplier terminations, the sign of selection effects is unclear ex ante. Larger customers may be faster to sever suppliers with more resources to assess climate exposure and higher bargaining power. At the same time, smaller customers could have less diversified supplier networks and react more strongly to perceived changes in exposure. The direct effect and propagation of the effects of adverse weather could be magnified for smaller firms, as previous studies show stronger effects of weather on output in less developed economies ([Burke et al. 2015](#)) and because private firms could be more strongly affected due to higher geographic concentrations and financial constraints. Hence, the first-stage estimates could be a lower bound.

locations. In additional analyses, we estimate the propagation of incidents aggregated across facilities.

1.3 Local temperatures and projections

The exposure to heat is projected to gradually increase in frequency and severity in the near future (CSSR 2017). We construct measures of firms' exposure to heat at the firm-quarter-level using location-specific information from the ERA5 reanalysis data provided by the European Center for Medium-term Weather Forecasts (ECMWF). Re-analyses are generated by interpolating local records using atmospheric models. ERA5 provides global, daily maximum temperatures on a 0.25° grid starting in 1979 (Hersbach et al. 2020). We match customers and suppliers to temperatures at the closest ERA5 grid node, and convert temperatures from Kelvin to Celsius. Accounting for firms' reporting schedules, we sum the number of days of heat per financial quarter or year. For the main measure, we use a daily maximum temperature of 30°C as the temperature threshold to define a day as hot. This choice is based on the literature on physiology, temperatures, and labor productivity (e.g., Graff-Zivin and Neidell (2014), Burke et al. (2015), Carleton and Hsiang (2016), Sepannen et al. (2006)), which documents a sharp decline in worker performance for temperatures above 30°C. Similarly, the National Weather Service defines heatwaves based on a sequence of days with temperatures exceeding 90°F (32°C). In robustness tests, we combine this absolute threshold with relative definitions of high temperatures based on day of the year- and location-specific temperature distributions from 1979 to 1999.

Next, we add temperature projections from the CMIP5 provided by the ECMWF. The CMIP5 is used in the Intergovernmental Panel on Climate Change (IPCC) Assessment Reports and described by Hurrell et al. (2011) and Taylor et al. (2012). To compare realized temperatures with projections, we calculate the projected change at supplier locations as the number of days over 30° C modeled from 2006 to 2019 to projections for 2040 to 2059 based on the output of the MPI-ESM-LR model, averaged across all available ensemble members. We obtain projections following the Representative Concentration Pathway (RCP) 2.6, 4.5, and 8.5, which provide temperature projections for different levels of future emission. The RCP 8.5 comes closest to a “business as usual scenario,” assuming limited efforts to reduce emissions.

In robustness tests, we compare our data with records of natural disasters from EM-DAT from the Centre for Research on the Epidemiology of Disasters (CRED 2011). The data are at the country-level. Hence, it is less clear whether firms were directly affected. EM-DAT is commonly used for international studies in the economic literature on natural disasters (e.g., Strömberg 2007; Noy 2009; Lesk et al. 2016). Table 1f shows that less than 6% (4%) of our observations coincide with records of fires (droughts) in the suppliers' home countries.

Table 1
Summary statistics

(a) Unique supplier-year-quarter observations						
	N	Mean	SD	p25	Median	p75
Total assets (US\$ billions)	202,439	3.145	13.214	0.114	0.399	1.457
Revenue/Assets (%)	202,439	26.660	20.664	12.224	21.859	35.221
Operating income/Assets (%)	202,439	1.020	4.664	−0.037	1.414	3.086
Heat days (30° C)	202,439	14.091	23.624	0.000	1.000	18.000
Heat days (within-location)	202,439	−0.000	16.170	−8.967	−1.128	3.571
Heat days (30° C-95P)	202,439	3.146	5.788	0.000	0.000	4.000
Heatwave (30° C/7 days)	202,439	0.243	0.429	0.000	0.000	0.000
Heatwave (30° C-95P/7 days)	202,439	0.032	0.175	0.000	0.000	0.000
Average temperature	202,439	19.292	5.591	15.016	19.197	22.325
Heat days (EM-DAT)	202,439	0.663	4.129	0.000	0.000	0.000
(b) Unique customer-year-quarter observations						
	N	Mean	SD	p25	Median	p75
Total assets (US\$ billions)	124,140	18.533	45.177	0.883	3.284	12.279
Market capitalization (US\$ billions)	123,379	9.166	16.117	0.675	2.608	9.140
Revenue/Assets (%)	125,359	25.757	20.339	11.788	21.137	33.914
Operating income/Assets (%)	125,359	1.802	3.132	0.472	1.665	3.203
(c) Unique firm-pair-year observations						
	N	Mean	StDev	p25	p50	p75
1 (Last Relationship Year)	131817	0.270	0.444	0.000	0.000	1.000
Relationship Age (Years)	131207	5.079	3.870	2.000	4.000	7.000
Pct. Sales Sup (%)	13924	19.382	18.141	10.000	14.000	23.000
Pct. COGS Cus (%)	11852	2.358	7.370	0.072	0.297	1.288
Sup-Cus HQ Distance	130964	5544.352	4091.436	1707.374	4305.921	8815.658
MCap Cus / MCap Sup	109036	223.363	457.354	4.665	29.512	165.017
Assets Cus / Assets Sup	110817	312.119	614.817	7.074	43.905	249.268
Sales Correlation	81465	0.149	0.425	−0.160	0.170	0.476
1 (Realized > Exp.) Heat Days (> 1)	131817	0.399	0.490	0.000	0.000	1.000
1 (Realized > Exp.) Heat Days	131817	0.513	0.500	0.000	1.000	1.000
RCP2.6 δ Heat Days 2006/19-2040/59	131811	7.026	8.385	2.186	6.202	9.833
RCP4.5 δ Heat Days 2006/19-2040/59	131811	11.781	13.375	3.071	9.462	15.124
RCP8.5 δ Heat Days 2006/19-2040/59	131811	22.046	18.131	9.431	21.507	27.133
(d) Unique supplier-year observations						
	N	Mean	StDev	p25	p50	p75
Asset Tangibility	22457	0.259	0.243	0.062	0.175	0.396
Ind. Vulnerability	23951	0.067	0.249	0.000	0.000	0.000
No. of Customers	23927	6.901	8.015	2.000	4.000	9.000
Ind. Competitiveness	23949	0.832	0.656	0.323	0.601	1.318
Ind. HHI MCap	23949	0.106	0.108	0.027	0.060	0.192
(e) Unique customer-year observations						
	N	Mean	StDev	p25	p50	p75
BEA Input-Ind. Concentration	14262	0.051	0.107	0.011	0.022	0.040
Supplier Diversification	26576	1.783	1.272	1.000	1.286	2.000
No. Suppliers	29278	9.024	19.847	1.000	3.000	8.000
No. Suppliers / Assets	26894	5.603	26.951	0.503	1.189	3.009
Acct. Payable / Assets	23223	0.109	0.101	0.039	0.078	0.143
Acct. Receivable / Assets	23340	0.164	0.127	0.069	0.134	0.224
COGS / Assets	23253	0.669	0.561	0.241	0.526	0.913
Inventory / Assets	23286	0.116	0.114	0.021	0.088	0.174

(Continued)

Table 1
(Continued)

(f) Unique supplier-year-quarter observations						
	N	Mean	StDev	p25	p50	p75
Supplier Heat Days (t)	202439	14.091	23.624	0.000	1.000	18.000
Heat Days ex. EM-DAT Heatwave	202439	13.559	23.215	0.000	0.000	17.000
Heat Days ex. EM-DAT Fire	202439	13.296	23.267	0.000	0.000	16.000
Heat Days ex. EM-DAT Drought	202439	13.960	23.584	0.000	1.000	18.000
Heat Days ex. EM-DAT Heatwave/Drought/Fire	202439	12.635	22.792	0.000	0.000	14.000

This table presents summary statistics at the quarterly level for suppliers (a,e) and customers (b) as well as at the yearly level for customer-supplier pairs (c), suppliers (d), and customers (e) in our sample. The sample period is 2003 to 2016. Table A.1 in the appendix defines all variables. Firm-level accounting data are from Worldscope. “Asset tangibility” is the ratio of PPE to total assets. “Ind. vulnerability” is an indicator that takes the value of one if the firm operates in a high climate-risk industry. “Industry Competitiveness” is the number of firms (in thousands) per SIC two-digit industry. “BEA Input-Ind. Concentration” is the Herfindahl-Hirschman index (HHI) of inputs per industry from the BEA Input-Output matrices. “Supplier diversification” is the ratio of the number of suppliers to unique supplier SIC two-digit industries. “Sales correlation” is the running correlation of supplier and customer sales over the previous 9 quarters. The number of suppliers and percentage of sales (COGS) are obtained from FactSet Revere. The sample excludes firms that operate in the financial industry (SIC one-digit code of 6).

Table 1 reports descriptive statistics with variable descriptions in Table A.1. The average supplier firm in our sample experiences 14 days above 30° C per financial quarter with an annual average temperature of 19° C (Table 1a). As in prior research (e.g., Banerjee et al. 2008), we find asymmetries between suppliers and customers (Table 1b). The median customer holds 12 times the assets of the median supplier. When sales volumes are disclosed (<10% of the sample), customers represent 19% of suppliers’ sales, whereas sales from suppliers only account for 2% of customers’ cost of goods sold (Table 1c). Measures of supplier and customer industry competitiveness are described in Table 1d and 1e. Table 1f compares our data with records of natural disasters in EM-DAT.

2. Heat, Supply Chains, and Financial Incentives to Adapt

Our main research question on the adaptation of supply chains implicitly assumes that the occurrence of heat at supplier firms has direct and indirect adverse effects on suppliers and customers, providing an incentive for customer firms to adapt when temperatures increase. To validate this assumption, we examine the direct adverse effects of weather on supplier performance and the downstream propagation effects on customers. The downstream effects of adverse weather in production networks are theoretically ambiguous. On the one hand, customers might mitigate the effects, that is through multisourcing or inventory management, so that distortions dissipate in the supply chain. Similarly, suppliers’ limited bargaining power could force them to absorb higher costs. On the other hand, even small disruptions could decrease production output and cause supply-chain glitches in modern just-in-time production, when the specificity of inputs is high (Barrot and Sauvagnat 2016), or when customers’ ability to procure inputs from alternative sources is limited.

2.1 Empirical strategy

Our validation exercise proceeds in two stages. The methodology follows [Barrot and Sauvagnat \(2016\)](#) and [Carvalho et al. \(2021\)](#), who document the direct effects and shock propagation from suppliers to customers following large-scale natural disasters. In contrast, we focus on adverse weather, which is projected to heterogeneously increase in frequency and severity around the world. In the first (second) test, we focus on supplier (customer) operating income scaled by assets as our main measure of performance. We lag assets by one year to ensure that potential direct effects of heat exposure on assets does not confound the results. Since adverse weather might distort suppliers' operations in ways not reflected in financial performance, the estimates could understate the extent to which supplier-customer relationships are challenged by high temperatures.

In both stages, we use variation in heat exposure at supplier locations, which is plausibly exogenous and randomly distributed *conditional* on locations and seasons. We isolate this variation through OLS regressions with firm-by-fiscal-quarter fixed effects. The specification serves two goals: First, weather and financial performance are likely endogenous in the cross-section, as firms select into locations based on various firm-level characteristics. In contrast, within-location variation in heat days is arguably exogenous, and can only be predicted with precision over very short horizons. Second, firm-by-fiscal-quarter fixed effects mitigate concerns about confounding firm-specific patterns of seasonality. For example, certain types of firms earn higher revenues in summer, which could correlate with the incidence of heat. Since seasonality differs between firms, firm-by-fiscal-quarter fixed effects address the issue better than individual firm and quarter fixed effects. In the second stage, we aggregate the quasi-random variation in exposure at supplier locations by customer-year-quarter (similar for example to [Kale and Shahrur 2007](#); [Banerjee et al. 2008](#); [Barrot and Sauvagnat 2016](#); [Campello and Gao 2017](#); [Cen et al. 2017](#); [Phua et al. 2018](#)). As it is ex ante unclear whether the effects manifest immediately, the main tests include three lags of heat days.

In both stages, we include industry-by-year-by-quarter fixed effects to absorb industry-specific time trends. We also include country-specific linear trends to control for simultaneous trends in temperatures and economic development. In line with [Barrot and Sauvagnat \(2016\)](#), we sort firms into size, age, and profitability terciles to introduce additional size-, age-, and profitability-specific time fixed effects to absorb common variation of different firm profiles. The implementation of the tests on direct and indirect effects is illustrated below. First, to examine the direct effects of heat days on suppliers, we estimate the following regression at the quarterly frequency,

$$y_{it} = \sum_{t=-3}^0 \beta_t \times W_{it} + \mu_{iq} + \gamma_{n(i)t} + \theta_{d(i)t} + \bar{\delta}_{BS2016_{it}} + \epsilon_{it} \quad (1)$$

where y_{it} is *Operating Income/Assets* of supplier i in year-quarter t . Adverse weather, W_{it} , is the number of days on which firm i was exposed to heat in t . Supplier-by-fiscal-quarter fixed effects are denoted by μ_{iq} , $\gamma_{n(i)t}$ are industry n of firm i by year-quarter t fixed effects based on two-digit SIC codes, $\theta_{d(i)t}$ are quarterly country d linear trends, and $\bar{\delta}_{BS2016_{it}}$ are firm size, age, and profitability terciles interacted with year-quarter fixed effects. Second, we estimate whether heat days indirectly affect customers with the regression

$$y_{ct} = \sum_{t=-3}^0 \beta_t \times W_{ct} + \mu_{cq} + \gamma_{n(c)t} + \theta_{d(c)t} + \bar{\delta}_{BS2016_{it}} + \epsilon_{ct} \quad (2)$$

where y_{ct} is *Operating Income/Assets* of customer c in period t and W_{ct} is the sum of heat days across suppliers of customer c in period t . μ_{cq} are customer-by-fiscal-quarter fixed effects, $\gamma_{n(c)t}$ are industry n of customer c by year-quarter fixed effects, $\theta_{d(c)t}$ are country $d(c)$ -specific quarterly trends, and $\bar{\delta}_{BS2016_{it}}$ are customer size, age, and profitability by year-quarter fixed effects. To preempt that both supplier and customer are simultaneously affected by a given event, directly through local infrastructure or through simultaneous demand-side effects, we exclude all customers-supplier pairs with customers located within a 500-km radius of the affected supplier from our analysis. As in Barrot and Sauvagnat (2016), robust standard errors are clustered at the firm level.⁶

2.2 Results

Table 2a reports the results for the direct effects on suppliers. From column 1 to 4, we sequentially introduce fixed effects. Across all specifications, an additional hot day significantly decreases suppliers' quarterly *Operating Income/Assets*. In the most conservative specification (column 4), we find a reduction of 0.0087 percentage points across the current and last three quarters. This effect is economically meaningful. The standard deviation of affected days is 16.2, translating the estimate into an 13.8% decrease in operating income over assets for a one-standard-deviation increase in heat exposure. As in Barrot and Sauvagnat (2016), the direct effects persist in the directly affected and two subsequent quarters.⁷

Panel 2b shows the indirect effects on customers of affected suppliers. In line with evidence on the propagation of idiosyncratic shocks in supply chains,

⁶ Table IA2 shows the robustness of the results to alternative specifications.

⁷ We provide additional tests in the Internet Appendix. First, we estimate the effects for subsets of firms with different degrees of geographic concentration. The plots in Figure IA.1 shows increasing (though insignificant) magnitudes with firm-level geographic concentration. Second, we examine heterogeneity by industry in Table IA.3. For heat, we observe pronounced effects in agriculture, transportation, manufacturing, mining and construction, and services, roughly in line with previous work on crop yields, outdoors industries, and labor and capital productivity (see, e.g., Zhang et al. 2018; Sepannen et al. 2006; Somanathan et al. 2021; Burke and Emerick 2016).

Table 2
Physical exposure to heat and financial incentives to adapt

(a) Direct Effect on Supplier					(b) Propagation to Customer				
Sup Opl (t)					Cus Opl (t)				
	(1)	(2)	(3)	(4)		(1)	(2)	(3)	(4)
Supplier Heat Days (t)	-0.0039*** (-2.96)	-0.0047*** (-3.48)	-0.0032** (-2.38)	-0.0033** (-2.40)	Supplier Heat Days (t)	-0.0003*** (-4.422)	-0.0002*** (-2.921)	-0.0002*** (-2.683)	-0.0001* (-1.933)
Supplier Heat Days (t-1)	-0.0018 (-1.39)	-0.0035*** (-2.64)	-0.0025* (-1.83)	-0.0027* (-1.95)	Supplier Heat Days (t-1)	-0.0006*** (-6.553)	-0.0004*** (-4.225)	-0.0003*** (-4.010)	-0.0003*** (-3.196)
Supplier Heat Days (t-2)	-0.0039*** (-3.01)	-0.0044*** (-3.22)	-0.0031*** (-2.32)	-0.0027** (-2.02)	Supplier Heat Days (t-2)	-0.0003*** (-4.145)	-0.0001** (-2.129)	-0.0001* (-1.851)	-0.0001** (-2.025)
Supplier Heat Days (t-3)	-0.0024* (-1.77)	-0.0023* (-1.67)	-0.0015 (-1.05)	-0.0015 (-1.05)	Supplier Heat Days (t-3)	-0.0005*** (-7.214)	-0.0003*** (-3.760)	-0.0002*** (-3.380)	-0.0002*** (-3.092)
Firm × Fiscal-Qtr FE	Yes	Yes	Yes	Yes	Firm × Fiscal-Qtr FE	Yes	Yes	Yes	Yes
Ind × Year-Qtr FE	No	Yes	Yes	Yes	Ind × Year-Qtr FE	No	Yes	Yes	Yes
Ctry-Linear-Trends	No	No	Yes	Yes	Ctry-Linear-Trends	No	No	Yes	Yes
BS2016 FE	No	No	No	Yes	BS2016 FE	No	No	No	Yes
Observations	202438	202438	202438	202438	Observations	123700	123700	123700	123700
Customers	5628	5628	5628	5628	Customers	6299	6299	6299	6299
R ²	.19	.623	.626	.631	R ²	.69	.706	.707	.711

This table presents OLS regression estimates on the effects of heat at the location of the sample supplier firms on the operating income (Opl), scaled by lagged assets, of suppliers (panel 2a) and their downstream customers (panel 2b). The dependent variable is multiplied by 100 for ease of interpretation. *Heat days (t)* indicate the number of hot days at the supplier firm during the financial quarter *t* and the three preceding quarters (*t* − 3 to *t* − 1). The number of observations refers to supplier firm year-quarters (panel 2a) and customer firm year-quarters (panel 2b). In panel 2b, we aggregate the number of supplier heat days for each customer year-quarter. The sample period is 2003 to 2016. We exclude firms in the financial industry as well as firms with less than 10% of firm locations within 30 kilometers of the company headquarter. The regressions include firm-by-fiscal quarter fixed effects to control for time invariant firm characteristics and firm-specific seasonal effects, industry-by-year-by-quarter fixed effects, controls for country-specific linear trends, and interaction terms of firm size, age, and ROA with year-by-quarter fixed effects to control for firm characteristics following Barrot and Sauvagnat (2016) as indicated. Standard errors are clustered at the firm level. * $p < .1$; ** $p < .05$; *** $p < .01$.

heat at the locations of the suppliers decreases the financial performance of downstream customers. Specifically, one additional day of heat reduces quarterly operating income over assets of the indirectly affected customer by 0.0007 percentage points summed over the lags $t = -4, \dots, 0$ (column 4). For a one-standard-deviation increase in heat days, the downstream effect translates into a 0.6% decrease in operating income over assets, corresponding to an absolute amount of $\approx 355,000$ US\$ for the average firm. The indirect effects are smaller than the direct effects at approximately 4.6% of the direct effect. For interpreting the magnitudes, it is important to note that these effects are estimated net of mitigating measures firms may already have put in place. Hence, the estimates are a lower bound, and likely to constitute financial incentives for customers to monitor potential increases in frequency and severity.

We provide additional analyses in the appendix. First, we estimate a dynamic version of Equation (2), using indicator variables to capture the effect on customer operating income in the $t - 4$ to $t + 6$ quarters around the occurrence of a heatwave at the supplier firm. The corresponding dynamics plot is shown in Figure A.1. While the coefficient estimates decline between periods $t - 3$ and $t - 1$, all coefficients in the pre-period are statistically indistinguishable from zero, at both the 5% (shown in the figure) and the 10% levels. Some of the decline may be due to imprecise measurement of the beginning of a heatwave across fiscal quarters. The coefficients drop significantly below zero in quarters $t + 1$ and $t + 2$, and subsequently gradually return to pre-event levels. Second, we replace our definition of hot days ($> 30^\circ \text{C}$) in Table A.2. We find similar effects for temperature thresholds that take the day of the year- and location-specific distributions of temperatures into account, as well as for indicators of heatwaves. Third, we measure heat exposure at the supplier location level using 1.1 million addresses of sites, branches, and subsidiaries from Orbis. Although this data does not include information on the nature, scope, and time variation in activity across locations, Table A.3a shows consistent significantly negative estimates. To rule out that the result is mechanical, we reestimate the tests on a sample of years during which the relationships between customers and suppliers are reportedly inactive. With the exception of one weakly positive coefficient, the results in Table A.3b show insignificant estimates. Fourth, we describe cross-sectional differences in Table A.4. Consistent with the literature, we find significantly stronger effects of heat for suppliers in the agricultural, mining, and construction sectors. We also find a significant moderating effect of industry competitiveness and an exacerbating effect of customer reliance on supplier inputs on the propagation of the effects of hot days, in line with the idea that customers' ability to switch suppliers or low supplier bargaining power mitigate adverse effects. Fifth, we examine other firm-level outcomes in Table A.5. We find decreases in revenues over assets, revenues over employees, revenue growth, operating margins, purchase volumes (customer accounts payable and COGS) and customer inventory.

To conclude these motivating tests, we assess our results in the context of Barrot and Sauvagnat (2016) and Carvalho et al. (2021). For the tests, we adjust heat days for the number of days on which suppliers' countries were exposed to natural disasters recorded in EM-DAT – heatwaves, droughts, and fires. In Table A.6a, we find similar results in magnitude and significance when we estimate the effects for heat days which did not coincide with disasters. Table A.6b shows the indirect effects on customer operating performance. Again, excluding heat-related disasters does not affect the magnitude or significance of the results. This finding points to a key difference between heat and other types of climate-change-related hazards: heat is much less salient, and may affect firm performance through mechanisms that are unrelated to large-scale destruction.

3. Supply Chain Adaptation

If adverse weather has financially material consequences and gradually increases in frequency, firms may face incentives to adapt their production networks. In this section, we describe how local trends in the frequency of heat days could affect firms' decisions to continue or terminate existing supply-chain relationships. To guide our empirical analysis, we first summarize the conceptual framework.⁸ Subsequently, we estimate the magnitude of firms' responses to perceived increases in climate change exposure based on observable supply-chain adjustments.

In a general setting of firm production, customers seek to maximize profits and carefully trade off various supplier characteristics before entering supply-chain relationships. These characteristics include environmental exposure, as adverse weather can affect supplier *and* downstream customer performance even at small magnitudes (see Section 2). Customers and suppliers match in equilibrium and prices and quantities are such that customers are unwilling to accept increases in exposure, all else equal. Therefore, changes in climate change exposure could lead customers to push for changes in prices, adjust quantities, take out insurance, or require additional contract provisions before terminating relationships. As we do not observe any of these terms, we focus on customers' decisions to switch suppliers in reduced form analyses.

Customers may obtain information on heat exposure from a range of sources. For example, managers may learn about past disruptions from the suppliers' disclosures and internal research. Similarly, heatwaves, fires, and droughts may be discussed by regional, national, or international outlets. Following Kelly et al. (2005), we assume that customers form priors about the distribution of supplier heat days using historical information. With climate change, the parameters of these distributions may change over time. However, customers

⁸ Internet Appendix Section IA.I shows a detailed version of the conceptual framework and the underlying assumptions.

cannot directly observe these changes and assess their suppliers' exposure in every period during the supply-chain relationship. If customers perceive increases in heat exposure beyond ex ante expectations, a previously optimal match with a supplier may no longer be optimal from the perspective of the customer. In contrast, if the likelihood of heat days at the supplier location is unchanged or decreases compared to what customers could have expected ex ante, customers have no incentive to reassess their supplier choice.

Previous research in finance and economics has proposed experience-based Bayesian learning to model how economic agents process information about changing environments (e.g., Alevy et al. 2007; Chiang et al. 2011). Under this framework, firms observe recent climate realizations and attempt to infer from *experienced* gradual changes whether the underlying distribution has shifted. This approach is not without difficulty: In the short- and medium-term, weather outcomes provide a noisy signal of potential changes in climate distributions. Nevertheless, empirical studies have documented that individuals exhibit behavior consistent with Bayesian learning when it comes to climate change (Deryugina 2013; Moore 2017; Kelly et al. 2005; Kala 2019; Choi et al. 2020).

Alternatively, firms could use forward-looking projections to assess expected changes in temperatures, and interpolate expectations for future periods based on current and projected probabilities of heat days (Moore 2017). As outlined by Fiedler et al. (2021), this approach is impractical for most businesses, as differences between scientific climate models are challenging to interpret, as projections are usually designed for long-term horizons and model outputs may not be sufficiently granular in spatial terms. Hence, our main tests focus on firms' responses to experienced changes. In additional analyses, we consider the role of long-term climate change projections.

3.1 Empirical strategy

We measure if realizations of adverse weather have increased beyond a proxy of customers' ex ante expectations as shown in Figure 2. Based on the conceptual framework, we assume that customers form a prior (*Expected Exposure*_{sct}) using historical records of the number of affected days per year at the supplier location *before* the start of any given relationship. Since it is unclear which period customers use for this purpose, we conduct robustness tests with different horizons of 5, 10, and 15 years. Starting in the beginning of the relationship, customers observe suppliers' exposure in every year of the relationship and assess whether the average realized number of heat days appears to be consistent with their ex ante expectation. Our main measure, $\mathbb{1}(\text{Realized} > \text{Expected Exposure})_{sct}$, takes the value of one in year t if the average number of heat days per year since the beginning of the relationship exceeds the corresponding prior for supplier s , and zero otherwise. In additional tests, we use the continuous difference, that is *Realized* – *Expected Exposure*_{sct}. Our main outcome variable is the indicator $\mathbb{1}(\text{End})_{sct}$

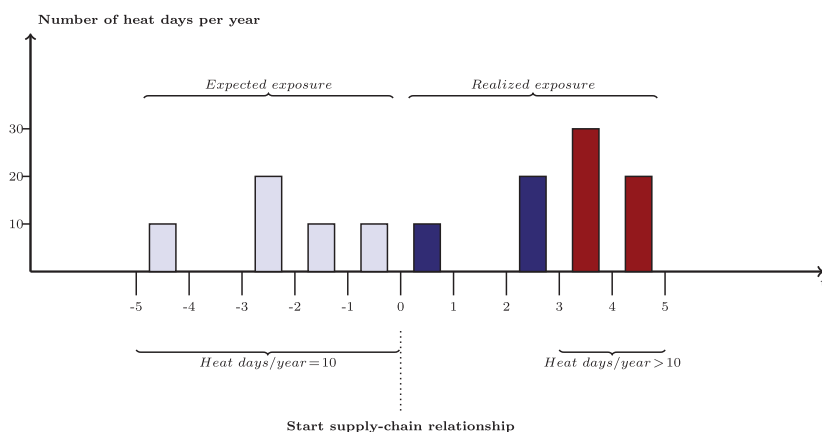


Figure 2

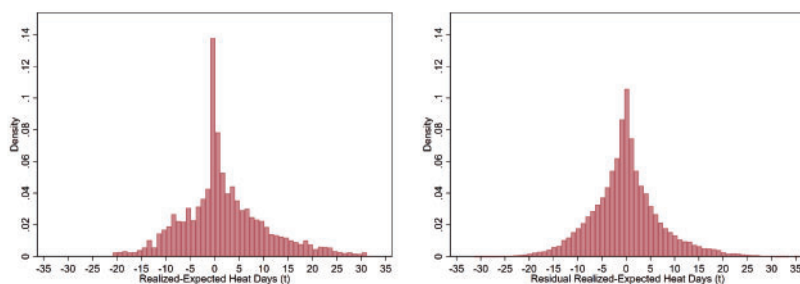
Variable construction: Expected versus realized exposure

This figure illustrates the construction of our main measure, $\mathbb{1}(\text{Realized} > \text{Expected Exposure})(t)$, capturing the discrepancy between realized and expected exposure of a hypothetical supplier to heat over time. This indicator variable is constructed by first estimating the historical prior as the average number of heat days per year in the supplier location over a benchmark period of 10 years *before* the establishment of a given supplier-customer relationship. In robustness tests we use alternative estimation horizons of 5, 7, and 15 years. $\mathbb{1}(\text{Realized} > \text{Expected exposure})(t)$ then takes the value of one in a given year t if the difference between the realized number of heat days per year since the beginning of the supplier-customer relationship exceeds the corresponding expected number of days, and zero otherwise. For example, in the case illustrated above, the average number of heat days at the supplier location over the 5-year benchmark period before the beginning of the supply-chain relationship is 10 (transparent bars). In year 1, the realized number of days is 10, not exceeding the expected value from the benchmark period. In year 3, the number of hot days is 20, bringing the average number of annual shocks since the beginning of the supply-chain relationship to 10 (i.e., $(10+0+20)/3=10$). Hence, the prior is still not exceeded (dark-blue-bars). In years 4 and 5, the average number of realized annual shocks increases above 10, exceeding the expected value from the benchmark period (bars in red). Thus, $\mathbb{1}(\text{Realized} > \text{Expected exposure})(t)$ would be zero in $t=1$ through $t=3$, and one in $t=4$ and $t=5$ in the illustrated example.

which is set to one in the last year of supply-chain relationships or zero if the relationships stay active.

Our identification strategy is similar to the long-differences approach by [Burke and Emerick \(2016\)](#), and relies on the fact that short-run climate realizations, in contrast to long-run changes, are randomly assigned across space. Intuitively, our empirical design leverages the idea that managers can incorporate expected levels of exposure, but not deviations from expectations into their decision-making. To provide supporting evidence for this identifying assumption, Figure 3 plots the distribution of the difference between *Realized* and *Expected exposure*, as well as the residual variation after we absorb high dimensional time-varying regional fixed effects. The distribution is largely unaffected by including fixed effects, in line with the idea that the underlying short-term trends are randomly assigned and not determined at the country and/or year levels. Based on this reasoning, our main tests estimate the following linear probability model at the annual frequency:

$$\begin{aligned} \mathbb{1}(\text{End})_{sct} = & \beta \times \mathbb{1}(\text{Realized} > \text{Expected Exposure})_{sct} \\ & + \gamma_{n(s)t} + \gamma_{n(c)t} + \theta_{d(s)d(c)t} + \epsilon_{sct} \end{aligned} \quad (3)$$



Identifying variation: Difference of realized and expected heat days

Figure 3

Identifying variation in realized vs. expected exposure to heat

This figure shows the distribution of our main measure, (*Realized > Expected Exposure*) (t), capturing the discrepancy between realized and expected heat days both before and during the supply-chain relationship. The construction of the variable is illustrated in Figure 2. First, we estimate the historical prior as the expected number of heat days per year in the supplier location over a benchmark period of 10 (in robustness tests 5, 7, and 15) years *before* the establishment of a given supplier-customer relationship. Second, we calculate the difference between this prior and the average realized number of heat days per year since the beginning of the supplier-customer relationship. The figure shows the distribution of the absolute deviation of realized and expected exposure to heat (*left*) as well as the residual variation after absorbing customer industry-year, supplier industry-year, and customer country-supplier country-year fixed effects (*right*).

To further control for potential confounding effects which coincidentally correlate with both climate trends and other reasons for relationship terminations, we estimate this model with fixed effects. First, we include supplier (customer) industry-by-year fixed effects, $\gamma_{n(s)t}$ ($\gamma_{n(c)t}$) to account for industry trends, for example, trends in make-or-buy choices. Second, we add supplier-country by customer-country by year fixed effects $\theta_{d(s)d(c)t}$ to account for changes in macroeconomic conditions, trade barriers, or import-related costs. Through this design, our results are identified with within-country-year and within-industry-year variation, making unobserved country- or industry-level shocks less likely to explain the results. Unobservable characteristics would have to systematically change in those firm-pairs where the realized exposure exceeds expectations, in a way that is unrelated to industry- or country-level trends. We do not include relationship or firm fixed effects in this specification, as our main independent variable captures within-relationship differences. To support the exclusion restriction that the observed trends affect suppliers only, we exclude all pairs with customers located within a 500-km radius. Tests with 1,000-km yield consistent results. In the literature on supply chains in finance (e.g., [Cen et al. 2018](#); [Chu et al. 2019](#); [Dasgupta et al. 2021](#); [Agca et al. 2022](#); [Chen et al. 2022](#)), it is common to cluster standard errors at the relationship level. In contrast, some recent papers on heat and economic outcomes note the potential for both spatial and temporal correlations and use two-way clustered standard errors (e.g., [Addoum et al. 2020](#); [Zhang et al. 2018](#)). From a conceptual point, clusters should reflect the level of treatment, which is defined by geographic and temporal factors in our setting. Yet,

Table 3
Expected versus realized exposure to heat and relationship termination

(a) Robust SE: Relationship clusters			
	1(Last Rel. Year)		
	(1) OLS	(2) Logit	(3) OLS
1(Real.>Exp. Heat Days)	1.013*** (3.982)		
1(Real.>Exp. Heat Days) (GLM)		0.0741*** (4.113)	
1(Real.>Exp. Heat Days) (> 1)			7.520*** (26.72)
Sup.-Ind.-Year FE	Yes	Yes	Yes
Cus.-Ind.-Year FE	Yes	Yes	Yes
Sup-Ctry-Cus-Ctry- Year FE	Yes	Yes	Yes
Observations	116,412	104,581	116,412
R ²	0.1290		0.1356
(b) Robust SE: Rel. and year clusters			
	1(Last Rel. Year)		
	(1) OLS	(2) Logit	(3) OLS
1(Real.>Exp. Heat Days)	1.013 (0.9493)		
1(Real.>Exp. Heat Days) (GLM)		0.0741 (0.9479)	
1(Real.>Exp. Heat Days) (> 1)			7.520*** (6.237)
Sup.-Ind.-Year FE	Yes	Yes	Yes
Cus.-Ind.-Year FE	Yes	Yes	Yes
Sup-Ctry-Cus-Ctry- Year FE	Yes	Yes	Yes
Observations	116,412	104,581	116,412
R ²	0.1290		0.1356

(Continued)

the implementation is not obvious: Spatial correlations of temperatures are unlikely to be well characterized by administrative boundaries, and temporally, temperatures may be clustered across or within years. Hence, both yearly and quarterly clusters do not entirely fit our treatment of within-country trends. As both too large or too small clusters can yield inaccurate errors (Verbeek 2021), we initially show specifications with and one- and two-way clustered standard errors and report the more conservative results in subsequent tests.

3.2 Results

Table 3 reports the results. Consistent with the hypothesis that existing suppliers are more likely to be terminated when the realized exposure to hot days exceeds customers' priors, we find positive and significant coefficients in Panel 3a. The linear probability estimates in column (1) indicate a one-percentage-point increase in the likelihood of supplier termination when realized heat days exceeded expectations. Based on the GLM fixed effects

Table 3
(Continued)

	(c) Magnitude of deviation		
	1 (Last Rel. Year)		
	(1)	(2)	(3)
1(Real.>Exp. Heat Days) = 1	-1.432 (-1.122)		
1(Real.>Exp. Heat Days) = 2	9.342*** (5.164)		
1(Real.>Exp. Heat Days) = 3	6.426*** (3.085)		
1(Real.>Exp. Heat Days) = 4	4.951* (2.037)		
1(Real.>Exp. Heat Days) = 5	0.9254 (0.7566)		
(Real.>Exp. Heat Days) (> 1)		0.1239 (1.719)	-0.0651 (-1.308)
(Real.>Exp. Heat Days) ² (> 1)			0.0269*** (7.171)
Sup.-Ctry.-Year FE	Yes	Yes	Yes
Sup.-Ind.-Year FE	Yes	Yes	Yes
Cus.-Ind.-Year FE	Yes	Yes	Yes
Sup.-Ctry.-Cus.-Ctry.-Year FE	Yes	Yes	Yes
Observations	116,412	115,298	115,298
R ²	0.1364	0.1290	0.1322

This table presents estimates of the effect of $1(\text{Realized} > \text{Expected})(t)$ on the likelihood of supply-chain relationship termination. The measure takes the value of one in year t if the difference between the average realized number of heat days per year since the beginning of the supply-chain relationship exceeds the expectation, and zero otherwise. Figure 2 illustrates the variable construction. The unit of observation is at the supplier-customer pair-year level. The dependent variable is a dummy variable taking the value of one if a given supplier-customer relationship ends after the current year t , and zero otherwise, multiplied by 100 for ease of interpretation. As in previous analyses, customer or supplier firms in the financial industry, supplier firms with less than 10% of locations within a radius of 30 km from the headquarters, and pairs with less than 500-km distance between headquarters are excluded from the sample. The regressions include year-by-supplier-country-, supplier-, and customer-industry-by-year, as well as supplier-country-by-customer-country-by-year fixed effects as indicated. Robust standard errors are clustered at the relationship level in panel A and double clustered at the relationship and year level in panels B and C. * $p < .1$; ** $p < .05$; *** $p < .01$.

logit regression in column (2), this effect translates into an increase of 7.4% in the likelihood of supplier termination. However, the effects are less precisely estimated in Panel 3b when we cluster standard errors more conservatively than in the supply-chain-finance literature. Assuming that customers assess average realizations of heat exposure relative to an ex ante expectation, the signals in the beginning of the relationship are particularly challenging to interpret. With quasi-mean varying weather realizations, the probability that realizations exceed expectations is close to 50% in the first year. Under experience-based learning, we would therefore expect weak responses to initial deviations. Indeed, when we set $1(\text{Realized} > \text{Expected Exposure})_{sct}$ to zero in the first year of the supply-chain relationship (> 1) in column (3), we find a larger and significant effect for both specifications (1% level).⁹

⁹ The robustness of the results under alternative approaches is shown in Table IA.5.

Moreover, we expect that observing repetitions of the signal increases customers' updating process, and that the likelihood of terminations increases with the number of periods during which the deviations persist. In Panel 3c, we estimate changes in the likelihood of supplier termination as a function of the number of years during which the realizations of adverse weather exceed customers' priors. In column 1, we estimate the effects of up to five repetitions, and impose no functional form on the marginal effect of each repetition. The effect of heat on supplier termination increases when expectations are exceeded for multiple years, and is particularly strong for the second signal. After 2 years, a prolonged observation of the deviation continues to increase the probability of supplier termination, but with a decreasing marginal effect. Under a learning process, we also expect that stronger observed deviations from expectations incur more pronounced updating (Deryugina 2013). We test for nonlinearities in firms' responses by estimating Equation (3) with the continuous measure of $(Realized - Expected)_{sct}$ and its square term in Panel 3c in column 2 and 3. We find that the likelihood of supplier terminations increases nonlinearly with the magnitude of the deviation.

We conduct two sets of robustness tests. First, we consider the duration of supplier-customer relationships as an alternative measure of supply-chain stability in Table A.7 similar to Fee et al. (2006) and Phua et al. (2018). We include strata for supplier- and customer-industry-by-year, and supplier-by-customer-country-by-year. In line with the main results, relationship duration decreases significantly when the realized number of heat days exceeds expectations. Second, our main test assumes that customers form priors over the 10-year period before the start of any given relationship. As Table A.8 shows, all estimates remain similar in magnitude and statistical significance when we estimate Equation (3) using alternative benchmark periods of 5 and 15 years.

The results align with the hypothesis that supply-chain managers form expectations about supplier heat exposure before entering the relationship, and compare expectations to realized weather throughout the relationship. However, the observed terminations could raise questions about how fast contracts can be cancelled when realizations exceed expectations. Supplier contracts can contain termination rights for cause or for convenience. Termination for convenience gives a contract party "the ability to terminate a contract at will, effectively creating an option to end the contract for any or no reason" (Thompson Hine LLP 2023) and contract parties with greater bargaining power may insist on contractual rights to end contracts "at any time for nothing more than convenience" (Foley & Lardner LLP 2023). Comprehensive data on contract provisions is not publicly available. Therefore, we hand-collect a sample of 27 supplier-contracts by searching the SEC EDGAR platform for "supply agreement." In line with the idea that contracts can be terminated on short notice, Table A.9 shows that more than 60% of contracts can be cancelled for convenience with an average termination notice

of about 120 days. Further, [Iyer and Sautner \(2018\)](#) study a sample of 185 proprietary contracts between a large transportation company and 89 suppliers. At an average contract term of 5.34 years, the observed switching time is 0.84 years.

An additional question related to adverse weather could be whether force majeure provisions protect suppliers from terminations. The precise definition of triggers is negotiated between contracting parties. However, it seems uncertain whether heat exposure would be considered severe enough, and heat days in our data do not generally overlap with country-wide heatwaves, droughts, or fires (Table 1f). Further, anecdotal evidence of court cases indicates that legal enforcement can be challenging for suppliers ([Dorsey & Whitney LLP 2023](#)). Specific to heat, the report cites the case of *Fru-Con v. United States*, in which a delay was not excused as the supplier “failed to show on which days, if any, excessive heat hindered or stopped critical work.” The report also states that proving “causation is particularly complicated when the force majeure event indirectly affects the contract such as, for example, disrupting shipping channels.”

3.3 Transitory heat exposure

Our conceptual framework implicitly introduces relationship terminations as customer decisions. While the previous results are consistent with the interpretation that customers perceive shifts in their suppliers’ exposure and extrapolate short-term trends of heat days, the observed terminations could be unrelated to customers’ changes in beliefs. For example, physical distortions, such as extended power outages, may disrupt the supply chain and require suppliers to abandon a contract. These two explanations are difficult to distinguish, both empirically and conceptually: it is plausible that customers are more likely to change their beliefs about long-term climate risks if they observe that weather events are affecting suppliers’ ability to fulfill the contract. In this case, the two interpretations of our result are not mutually exclusive but rather reinforce each other.

Nonetheless, to further document that our main finding is not driven entirely by short-term supplier-disruptions, we next study the effect of transitory weather shocks on the likelihood of supplier termination. If supply-side disruptions caused by high temperatures fully explained our results, we would expect the effect of transitory shocks on supplier termination to be of a similar magnitude as the exceedance of ex-ante expectations, documented in Table 3. We test this notion using within-supplier location variation in *Realized Heat Days*, which measures adverse realizations of heat regardless of expectations from a benchmark period.

The results, summarized in Table 4, show an economically small and statistically insignificant contemporaneous effect of heat on supply-chain termination across all specifications. These insignificant estimates are inconsistent with the hypothesis that customers primarily respond to salient, transitory shocks when

Table 4
Relationship termination in response to transitory heat exposure

	OLS - 1 (Last Relationship Year)				
	(1)	(2)	(3)	(4)	(5)
Realized Heat Days	0.0333 (1.561)	0.0235 (1.033)	0.0222 (0.9207)	0.0114 (0.4327)	0.0114 (0.4327)
Supplier Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Sup.-Ctry.-Year FE	No	Yes	Yes	Yes	Yes
Sup.-Ctry.-Cus.-Ctry.-Year FE	No	No	Yes	Yes	Yes
Sup.-Ind.-Year FE	No	No	No	Yes	Yes
Cus.-Ind.-Year FE	No	No	No	Yes	Yes
Observations	116,412	116,412	116,412	116,412	116,412
R ²	0.1639	0.2158	0.2618	0.2666	0.2666

This table presents linear probability model estimates on the effect of *transitory* heat occurrence on the likelihood of relationship termination. *Realized Heat Days* measures the number of hot days at the supplier locations in a given year. The unit of observation is at the supplier-customer pair-year level. The dependent variable is a dummy variable taking the value of one if a given supplier-customer relationship ends after the current year t , and zero otherwise, multiplied by 100. Customer or supplier firms in the financial industry, supplier firms with less than 10% of locations within a radius of 30 km from the headquarters, and pairs with less than 500-km distance between headquarters' are excluded from the tests. Fixed effects are included as indicated in the table. Robust standard errors are double clustered at the relationship and year level. * $p < .1$; ** $p < .05$; *** $p < .01$.

terminating existing supply-chain relationships. Instead, they are consistent with the idea that customers consider supplier environmental exposure when entering supply-chain relationships, as the occurrence of weather shocks in line with ex ante expectations does not meaningfully affect supplier terminations.

3.3.1 Other margins of adjustment If customers' expectations about suppliers' future exposure to climate hazards are changing, we might expect to see other adjustments in addition to the replacement of suppliers. For instance, customers may adjust their inventory management, invest in new production processes, or hold more cash to adapt to gradually deteriorating conditions. We test these hypotheses in Table 5.

For outcomes at the customer level, we collapse our panel by customer and year and aggregate the number of exceedances across suppliers. We find that when customers experience increases in adverse weather beyond expectations at their suppliers, inventories increase (column 1), potentially as a buffer for delayed deliveries. In line with the idea that adjustments of supply chains and larger inventories may be costly, we find an increase in the cost of goods sold (columns 2) and spending on research and development (column 3). Similar to Dessaint and Matray (2017), we also find an increase in cash holdings (column 4).

In Panel 5b, we test if customers adapt to perceived increases in climate change exposure by diversifying suppliers through dual sourcing. Also, customers could acquire suppliers to exert control over suppliers' adaptation efforts. Specifically, we study two outcomes: the number of suppliers in a relationship with the given customer operating in the same industry as the focal

Table 5
Other adjustments and supplier diversification

(a) Other margins of adjustment				
	Inventory (1)	COGS (2)	R&D (3)	Cash (4)
1(Real.>Exp. Heat Days)	0.0071* (2.095)	0.0092** (2.451)	0.0198*** (4.434)	0.0246*** (3.973)
Customer Firm FE	Yes	Yes	Yes	Yes
Cus.-Ind.-Year FE	Yes	Yes	Yes	Yes
Cus.-Ctry-Year FE	Yes	Yes	Yes	Yes
Observations	21,912	23,855	13,183	21,325
R ²	0.9774	0.9765	0.9755	0.8727
(b) Supplier diversification				
	Sup. SIC2 (1)	Sup. SIC3 (2)	Acq. SIC2 (3)	Acq. SIC3 (4)
1(Real.>Exp. Heat Days)	0.0552** (2.334)	0.0296 (1.521)	-0.0043 (-1.331)	-0.0026 (-0.9836)
Sup.-Ind.-Year FE	Yes	Yes	Yes	Yes
Cus.-Ind.-Year FE	Yes	Yes	Yes	Yes
Sup-Ctry-Cus-Ctry-Year FE	Yes	Yes	Yes	Yes
Observations	76,057	76,062	111,918	111,918
R ²	0.2105	0.1757	0.1095	0.1036

This table presents OLS estimates on the effect of supplier heat exposure on customer firm risk management and supply-chain diversification outcomes. In both Panels 5a and 5b, we aggregate observations of 1(*Realized > Expected Heat Days*) at the customer-year level to study customer-level outcomes. Panel 5a examines the effect on customer inventory, cost-of-goods-sold, R&D expenditures, and cash, (all log-transformed) as dependent variables in columns 1 through 8. Panel 5b examines the effect on supplier diversification and acquisitions. The dependent variable in columns 1 through 4 is the number of other suppliers in a relationship with the customer operating in the same SIC two- and three-digit industries as the focal supplier in the following year ($t+1$). The dependent variable in columns 5 through 8 is the number of acquisitions announced by the customer of firms in the SIC two- and three-digit industries of the focal supplier in the following year ($t+1$). Fixed effects are included as indicated in the table. Robust standard errors are double clustered at the customer and year level. * $p < .1$; ** $p < .05$; *** $p < .01$.

supplier, and the number of acquisitions of firms in the focal supplier's industry (SIC2 or SIC3). We find a positive effect of the exceedance of expected heat days on the number of suppliers in the same industry as the focal supplier but only for suppliers in the same two-digit, not in the same the three-digit SIC industry. We find no evidence for acquisitions of suppliers in the same industry as the focal supplier.

Apart from these margins, firms could move or change operations to accommodate changes in the environment. While we mostly study large and listed firms, we can neither rule out nor observe such changes. To the extent that business changes or relocations prevent terminations, we may underestimate the effects. Further, changes in location could incur noise in our measure of heat days and decrease the precision of the estimates.

3.3.2 Supply chain integration, industry reliance, and supplier competition To test the plausibility and explore the mechanisms behind our result, we

Table 6
Supply-chain integration, industry reliance, and supplier competition

	Dep. Var: 1(Last Relationship Year)						
	Cus.-Sup. Integration			Ind. Reliance		Sup. Competition	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1(Real.>Exp. Heat Days) (> 1)	16.47*** (3.525)	15.02*** (7.288)	10.35*** (6.980)	13.17** (2.874)	9.511*** (4.359)	5.432*** (3.420)	7.525*** (6.241)
1(Real.>Exp. Heat Days) (> 1) × Log(Sales to Cus)	-0.8276** (-2.932)						
1(Real.>Exp. Heat Days) (> 1) × Rel. Length		-2.838*** (-8.618)					
1(Real.>Exp. Heat Days) (> 1) × Sup. Diversification			-0.8350*** (-5.884)				
1(Real.>Exp. Heat Days) (> 1) × Sup.-to-Cus.-Ind. Sales				-15.14* (-1.843)			
1(Real.>Exp. Heat Days) (> 1) × HHI Sup.-Ind. Inputs					-6.619* (-2.113)		
1(Real.>Exp. Heat Days) (> 1) × N Firms Sup.-Ind.						1.594** (2.281)	
1(Real.>Exp. Heat Days) (> 1) × HHI Sales Sup.-Ind.							-0.0861*** (-6.145)
Log(Sales to Cus)	-1.592*** (-5.373)						
Rel. Length		1.487*** (8.915)					
Sup. Diversification			-0.5014*** (-6.077)				
Sup.-to-Cus.-Ind. Sales				-1.651 (-0.2919)			
HHI Sup.-Ind. Inputs					-2.229 (-0.9693)		
N Firms Sup.-Ind.						0.2698 (0.4884)	
HHI Sales Sup.-Ind.							-0.0846*** (-15.04)
Sup.-Ind.-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cus.-Ind.-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sup-Ctry-Cus-Ctry-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	18,557	116,412	102,222	17,777	47,393	108,511	116,407
R ²	0.2150	0.1426	0.1449	0.3782	0.2084	0.1436	0.1356

This table shows the cross-sectional heterogeneity in the effect of heat exposure exceeding ex ante expectations on the likelihood of supply-chain relationship termination. The measure 1(*Realized* > *Expected*) (> 1) is constructed as illustrated in Figure 2. The unit of observation is at the supplier-customer pair-year level. The dependent variable is a dummy variable taking the value of one if a given supplier-customer relationship ends after the current year *t*, and zero otherwise. “log(Sales to Cus)” is the log-transformed dollar value of sales from the supplier to the customer, “Rel. length” is the relationship length, that is, the number of years since the start of the relationship, and “Supplier diversification” is the number of suppliers, scaled by the number of unique supplier-industries for a given customer. “Supplier-industry to customer-industry sales” is the value of sales from the customer to the supplier industry, and “HHI sup.-ind. inputs” is the Herfindahl index of input industries per customer industry, both obtained from the BEA Input-Output matrices. “N firms sup.-ind.” is the number of firms in the supplier industry and “HHI sales sup.-ind.” is the Herfindahl index of sales within the supplier industry. We apply the same data filters as in Table 3. Each regression includes fixed effects as indicated. *t*-statistics in parentheses are based on robust standard errors double-clustered at the relationship and year level. **p* < .1; ***p* < .05; ****p* < .01.

examine cross-sectional differences of suppliers, customers, and relationships. In particular, we interact the main measure of perceived increases in heat days with proxies of customer-supplier integration, industry reliance, and supplier competition. Table 6 reports the results.

First, we study the role of supply-chain integration for customer adaptation to perceived changes in suppliers’ exposure. Customers typically make

relationship-specific investments, incurring switching costs. We collect three proxies of supply-chain integration: the value of supplier-customer sales, the length of the relationship, and the number of suppliers per industry as a proxy of customer dependence. Across the measures (column 1-3), higher integration or reliance on the supplier is related to weaker customers' reactions to experienced adverse weather in excess of expectations. Second, we consider the reliance of the customer on inputs at the industry level using BEA input-output matrices. In the cross-section, customers who are more reliant on their suppliers' inputs are less likely to terminate relationships in response to perceived increases in heat exposure (column 4 and 5). Third, we examine the role of competition in the supplier industry. If our findings are driven by customers substituting potentially risky suppliers, we would expect to find a stronger effect when competition in the supplier industry is high. As measures of competition, we use the number of firms in the SIC two-digit supplier industry and the HHI of sales in the supplier industry. The results are consistent with the conjecture. A one-standard-deviation increase in supplier-industry competitiveness more than doubles the effect of heat exposure exceedance on supplier termination, relative to the average effect (column 6 and 7).

3.3.3 Climate adaptation readiness While this paper focuses on customers' learning behavior and margins of adjustments, suppliers are likely to simultaneously engage in adaptive investments. Information on firms' private investments is difficult to obtain. Therefore, we use country-level data from the Notre Dame-Global Adaptation (ND-GAIN) Country Index to test if the observed effects are attenuated by higher levels of adaptation in supplier countries. ND-GAIN provides open-source information on countries' current vulnerability to climate disruptions, and assesses their readiness to leverage private and public sector investment. The overall readiness index is a weighted average of the three subcomponents: economic, governance, and social readiness. In this context, economic readiness is defined as "the investment climate that facilitates mobilizing capitals from private sector." Governance readiness is described as the "stability of the society and institutional arrangements that contribute to the investment risks." Social readiness constitutes "social conditions that help society to make efficient and equitable use of investment and yield more benefit from the investment."

Table 7 shows how perceived increases in supplier exposure and country-level readiness interact. ND-GAIN acknowledges that some aspects of their indexes overlap with measures of economic development. Hence, we are cautious in interpreting the effects. In line with the idea that suppliers in more adaptive countries could be less vulnerable to future heat events, we find a negative effect for the interaction of $1(Realized > Expected Exposure)_{set}$ with "Overall" and "Economic" adaptation readiness. "Overall" readiness is distributed on a scale of 0 to 1 with a mean of 0.64 and a standard deviation of 0.09. Thus, the coefficient indicates that the likelihood of supplier

Table 7
Climate change adaptation in the supplier country

(a) Supplier country adaptation readiness (ND-GAIN)				
	Dep. Var: 1(Last Relationship Year)			
	(1)	(2)	(3)	(4)
1(Real.>Exp. Heat Days) (> 1)	24.33*** (3.166)	29.19*** (3.072)	18.10** (2.824)	16.95** (2.903)
1(Real.>Exp. Heat Days) (> 1) × Overall	-23.25** (-2.183)			
1(Real.>Exp. Heat Days) (> 1) × Economic		-28.81** (-2.377)		
1(Real.>Exp. Heat Days) (> 1) × Governance			-13.24 (-1.528)	
1(Real.>Exp. Heat Days) (> 1) × Social				-14.17 (-1.470)
Sup.-Ind.-Year FE	Yes	Yes	Yes	Yes
Cus.-Ind.-Year FE	Yes	Yes	Yes	Yes
Sup-Ctry-Cus-Ctry-Year FE	Yes	Yes	Yes	Yes
Observations	109,242	109,228	109,208	109,242
R ²	0.1964	0.1966	0.1961	0.1963
(b) Country-level disaster declarations				
	Cus OpI (t)			
	(1)		(2)	
Heat Days (t,t-3)	-0.0002*** (-4.30)			
Heat Days No Declaration (t,t-3)			-0.0002*** (-4.24)	
Heat Days Declaration (t,t-3)			-0.0003*** (-3.90)	
Firm × Fiscal-Qtr FE	Yes		Yes	
Ind × Year-Qtr FE	Yes		Yes	
Ctry-Linear-Trends	Yes		Yes	
BS2016 FE	Yes		Yes	
Observations	123700		123700	
Customers	6299		6299	
R ²	.711		.711	

This table shows estimates on the cross-sectional heterogeneity of the effects related to country-level climate change adaptation (Table 7a) and disaster declarations (Table 7b) in the supplier country. “Overall,” “Economic,” “Governance,” and “Social” are indexes capturing the climate change adaptation readiness of the supplier country, obtained from the Notre Dame-Global Adaptation (ND-GAIN) Country Index. “Overall” is a weighted average of economic, governance, and social readiness for climate change adaptation. In Table 7a, the unit of observation is at the supplier-customer pair-year level. The dependent variable is a dummy variable taking the value of one, scaled by 100 for ease of interpretation, if a given supplier-customer relationship ends after the current year t , and zero otherwise. The specification and data filters follow Table 3. In Table 7b the unit of observation is at the customer-year-quarter level. The dependent variable is customer operating income scaled by customer assets lagged by a year and multiplied by 100. “(No) Declaration” means that there are (no) records of a declaration of a natural disaster in the same calendar quarter and home country of the supplier. We apply the same specifications and data filters as in Table 2. t -Statistics in parentheses are based on robust standard errors double-clustered at the relationship and year level. * $p < .1$; ** $p < .05$; *** $p < .01$.

terminations under perceived increases in heat is reduced by 8.6% for a one-standard deviation increase in readiness (column 1). In contrast, we do not find significant effects for “social” or “governance” readiness.

We investigate if the effects of economic readiness could be driven by transfer payments using data on disaster declarations and aid payments from

EM-DAT. In the sample period from 2003 to 2016, we find 5,586 records of natural disasters with 91 labeled as heatwaves. However, out of those 91, we find only four records of disaster declarations and no records of aid payments. Whereas the coverage is incomplete, the numbers support the idea that heat-related damages differ from other hazards in their salience. As Table 7b shows, we find significant evidence of the propagation of heat-related repercussions regardless of whether supplier countries declared an emergency or not. The difference of the coefficients is not statistically different from zero (p -value .103).

3.4 Experienced heat days and temperature projections

Our framework assumes that firms form expectations about their suppliers' exposure based on backward-looking information, and observe if average realizations deviate from their priors during the relationship. However, this approach is challenging, as weather outcomes are noisy signals of potential long-run change. To investigate possible limitations of firms' responses, we examine how firms respond when experienced changes do not align with future projections.

To do so, we focus on cases with limited projected change in local heat days until mid-century. If projections indicate minimal change going forward, relationship-specific investments to adopt new suppliers in response to short-term trends may be undesirable. We again estimate the regression in Equation (3) for subsamples with little projected change in long-term temperatures according to the RCP 2.6 (column 1), RCP 4.5 (column 2), and RCP 8.5 (column 3) scenarios. The three scenarios represent three of the main trajectories adopted by the IPCC, with the RCP 2.6 representing a very stringent scenario with strong policy intervention. RCP 8.5 is closest to a business as usual scenario, assuming very limited policy interventions directed at emissions reduction. Specifically, we focus on observations with a projected difference in heat days of less than 7 days per year comparing the periods 2006–2019 and 2049–2060.

Table 8 presents the estimates. Across all scenarios, the magnitude of responses to deviations of expected and experienced exposure are similar to our main results in Section 3. One possible interpretation of this result is that firms respond to experienced changes regardless of whether they coincide with projections. As Moore (2017) points out, adaptation will only occur at the speed at which economic agents learn about climate change. If investments are based on experienced increases, adaptation may insufficiently reflect long-term projections and progress slower than necessary.

4. Heat Exposure and Supplier Replacement

In the last part of our analysis, we examine how perceived increases in supplier exposure to heat affect the selection of new suppliers. The tests

Table 8
Experienced Exposure and Projections

	Dep. Var.: $\mathbb{1}(\text{Last Relationship Year})$			
	≈ 0 RCP2.6 (1)	≈ 0 RCP4.5 (2)	≈ 0 RCP8.5 (3)	Full Sample (4)
$\mathbb{1}(\text{Real.} > \text{Exp. Heat Days}) (> 1)$	7.595*** (3.390)	8.474** (3.008)	10.33** (3.032)	7.853*** (5.595)
Sup.-Ind.-Year FE	Yes	Yes	Yes	Yes
Cus.-Ind.-Year FE	Yes	Yes	Yes	Yes
Sup-Ctry-Cus-Ctry-Year FE	Yes	Yes	Yes	Yes
Observations	45,324	30,362	18,563	92,792
R^2	0.1715	0.2183	0.2563	0.1488

This table presents linear probability model estimates on the effect of heat exceedance, $\mathbb{1}(\text{Realized} > \text{Expected}) (> 1)$, on the likelihood of supply-chain relationship termination. Estimates are presented separately for the full sample (column 4), as well as subsets of suppliers located in areas which are projected to experience limited change in temperatures. The projected change is estimated as the difference between the number of days over 30° C from 2006-2019 and 2040-2049. Projections are obtained from the MPI-ESM-LR model, and averaged across all available ensemble members for the RCP 2.6, 4.5, and 8.5 scenario. We exclude observations before the issue of the IPCC 4th assessment report in 2007. The expected exposure is estimated over 10 years prior to the relationship. The unit of observation is at the supplier-customer pair-year level. The dependent variable is a dummy variable taking the value of one if a given supplier-customer relationship ends after the current year t , and zero otherwise. The variable is scaled by 100 for ease of interpretation. As in previous analyses, customer or supplier firms in the financial industry, supplier firms with less than 10% of locations within a radius of 30km from the headquarters, and pairs with less than 500km distance between headquarters' are excluded from the tests. The regressions include supplier and customer-industry-by-year fixed effects and supplier-country-by-customer-country-by-year fixed effects as indicated. Robust standard errors are double-clustered at the relationship and year level. * $p < .1$; ** $p < .05$; *** $p < .01$.

speak to two questions from the previous analyses: First, do firms deliberately manage supplier exposure to climate hazards? If customers observe financial repercussions but are agnostic about the driver, the exposure of old and new suppliers may remain similar. Second, how do firms assess noisy signals from short-run trends?

To address these questions, we look for replacement suppliers who enter a new supply-chain relationship with the same customer after observing an end of a relationship in our sample, as illustrated in Figure 4. We require replacement suppliers to have the same four-digit SIC code as the old supplier. As before, we drop customer-supplier pairs located in the same geographic region, and exclude customers and suppliers in the financial industry and firms with a concentration of facilities around the headquarters below 10%. Thereby, we identify replacements for 16,900 customer-supplier pairs in our sample.

We assume that customers evaluate suppliers and potential replacements during the initial supplier-relationship. However, if they ignore that short-run trends are noisy, replacement suppliers could have experienced more favorable conditions during the initial relationship despite similar distributions of adverse weather. Therefore, we have to compare exposures both *during* and *after* the initial relationship as in 4. For each period, we estimate the following linear probability model

$$\mathbb{1}(\text{Exposure New} < \text{Old})_{sc} = \beta \times \mathbb{1}(\text{Realized} > \text{Exp. Exposure})_{st} \tag{4}$$
$$+ \gamma_{n(s)t} + \gamma_{n(c)t} + \theta_{d(s)t} + \theta_{d(c)t} + \epsilon_{sc},$$

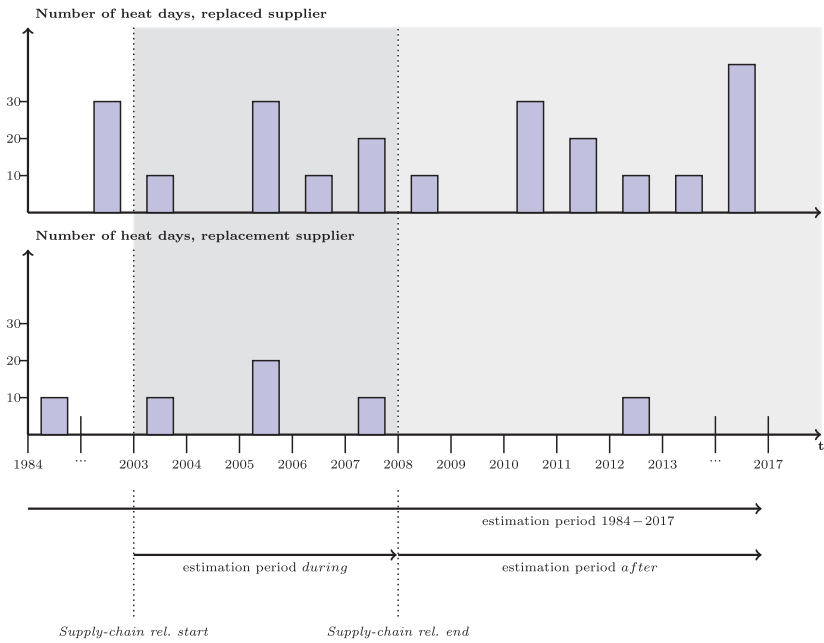


Figure 4
Variable construction: Exposure of replaced and replacement suppliers

This figure illustrates the construction of the comparison of the exposure to heat of replaced and replacement suppliers (Table 9) based on a hypothetical example of a supplier and the replacement. We compare the heat days of old and new suppliers based on three time periods. First, we estimate and compare the exposure of the replaced and replacement supplier based on the years (*in dark gray*) during which the initial supply-chain relationship was active. Second, we compare the exposure of both suppliers after the initial supplier has been replaced (*in light gray*). Third, we compare the exposure of both suppliers according to long-term projections of temperatures from the CMIP5 project.

where $\mathbb{1}(Exposure\ New < Old)_{sc}$ takes the value of one if the new supplier has a lower exposure to heat than the old supplier. $\mathbb{1}(Realized > Expected\ Exposure)_{st}$ indicates the exceedance of heat expectations at the location of supplier s in the initial relationship year t before termination to identify relationships that may have been terminated related to climate change considerations. We include customer and supplier industry- and (customer times supplier) country-by-year fixed effects (γ_{nt}, θ_{ct}).

Table 9 shows average effects, Figure 5 plots the corresponding distributions. We find a positive effect of the exceedance of ex ante expectations on the likelihood that replacement suppliers have a lower ex post exposure to heat than terminated suppliers *during* the initial relationship (columns 1 and 2). The likelihood that new suppliers had a lower heat exposure than old suppliers is 12-percentage-points higher for terminations after perceived increases ($\mathbb{1}(Realized > Expected\ exposure)=1$). Hence, customers might choose replacement suppliers which exhibited lower climate exposure in the

Table 9
Heat exposure and supplier substitution

	1(Decrease During Ini. Rel.)		1(Decrease After Ini. Rel.)		1(Decrease Projected Days)	
	(1)	(2)	(3)	(4)	(5)	(6)
1(Real.>Exp. Heat Days)	0.1508*** (7.535)	0.1229*** (7.635)	0.0947*** (4.562)	0.0560*** (2.702)	0.0543*** (2.864)	-0.0188 (-1.111)
Cus.-Ind.-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Sup.-Ind.-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Sup-Ctry-Cus-Ctry-Year FE	No	Yes	No	Yes	No	Yes
Observations	17,178	17,178	17,178	17,178	17,178	17,178
R ²	0.0771	0.2475	0.0672	0.2394	0.0560	0.2429

This table shows the effect of the exceedance of the expected physical exposure to heat on supplier substitution. We match supplier firms for which the supplier-customer relationship is terminated during the sample period (i.e., *old suppliers*) with their likely replacements (i.e., *new suppliers*). Replacements are identified as firms with identical four-digit SIC codes, which enter a new supply-chain relationship with the given customer within one year of the termination of the previous supply-chain relationship. The results show linear probability model estimates on the likelihood that the exposure to hot days of *new* replacement suppliers is lower than the exposure of *old* replaced suppliers as a function of 1(*Realized > Expected*). The dependent variable takes the value of one if new suppliers are exposed to fewer heat days than old suppliers, during (columns 1 and 2) and after (columns 3 and 4) the initial supply-chain relationship. In column 5 and 6, the dependent variable takes the value of one if fewer hot days are projected at the location of the new compared to the old supplier. Standard errors are double clustered at the relationship and first relationship-year level. * $p < .1$; ** $p < .05$; *** $p < .01$.

past when they conjecture gradual change materializes in the location of the old supplier.

At the same time, even if customers were unaware of observed changes in heat days and switched to replacement suppliers in ex ante similar climate zones, we might find a difference in climate exposure *during* the initial relationship by construction, as the old supplier experienced adverse outcomes drawn randomly from the underlying distribution. However, we would expect no difference in climate exposure between old and new suppliers *after* the initial relationship, with similar ex ante expectations for both firms. Importantly, a large proportion of the documented effect remains when we consider the period *after* the initial relationship ended. New suppliers were 6 to 9 percentage points more likely (significant at the 5% and 1% levels) to experience a decrease in heat exposure compared to old suppliers *going forward* (column 3 and 4). This result indicates that customers on average choose replacement suppliers with an ex ante different local temperature distribution.

We further consider long-term projections and use future heat days for 2040 to 2069 under the RCP 4.5 scenario modeled by the Max Planck Institute for Meteorology. We find a positive but less robust effect of the exceedance of expected heat days on the difference in projected heat between new and old suppliers (column 5 and 6). Taken together, increases in exposure that could be indicative of gradual climate change affect not only the termination but also the formation of new supply-chain relationships, as customers switch from suppliers which experienced worse conditions than expected to replacements in less exposed areas. However, the effects weakens when we consider long-term heat projections.

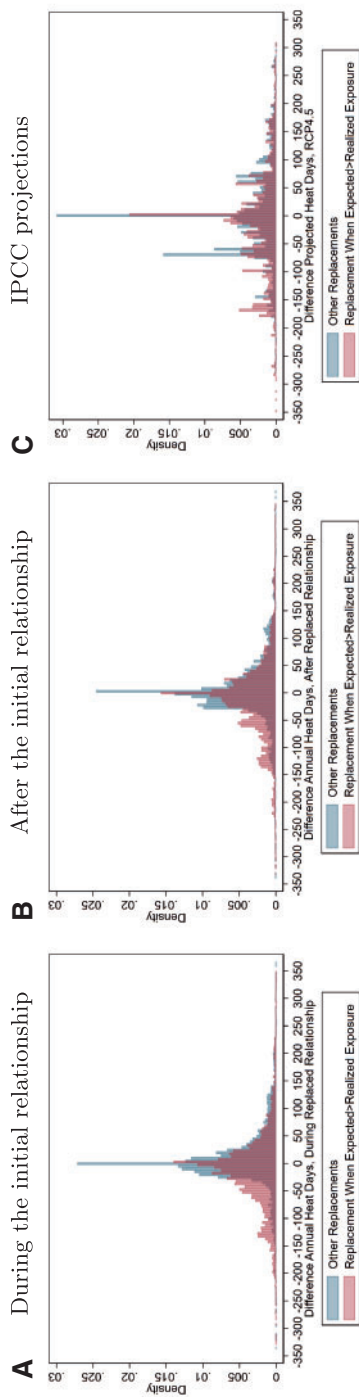


Figure 5

Heat exposure and supplier substitution

This figure shows the effect of an exceedance of ex ante expectations of heat exposure during the initial relationship on the exposure of new suppliers, analogous to Table 9. It plots the distribution of the difference in annual heat days between the original supplier and the potential replacement, identified based on four-digit SIC codes and a start date in the year in which the initial relationship is terminated. The figures separately plot the distribution for supplier replacement cases in which realized heat days exceeded expected heat days (red), as these terminations could have been related to climate-change-related considerations. The blue distribution plots all other supplier replacements. In terms of the affected time periods, the figures compare the exposure of initial and replacement suppliers during (Figure A) and after (Figure B) the initial supply-chain relationship, and compares projected exposures according to the CMIP5 RCP4.5 projections (Figure C).

5. Effects of Floods on Supply Chains

For a broader perspective, we replicate our tests with data on floods from the Dartmouth Flood Observatory based on satellite images, remote sensing sources, and news reports. We spatially match firms' headquarters and flooded areas and compute the number of flood days per financial quarter or year. On average, 6.5% of supplier-quarters are affected with floods lasting 10.7 days conditional on occurrence (Table IA.6). Like heat, floods decrease both the operating income over assets of the directly affected supplier and their customers (Table IA.7). In line with expectations, the magnitude of the effects is relatively larger. Further, the likelihood of relationship terminations increases when the realized number of flood days exceeds ex ante expectations (Table IA.8a). We also observe stronger responses with repeated exceedances (Table IA.8b). Despite the fact that floods can be destructive, the effects persist when we exclude physically affected suppliers, and transitory occurrences do not drive terminations (Table IA.9). Similar to heat exceedances, customers replace suppliers with less exposed new suppliers when we compare exposures during the initial relationship (Table IA.10). However, this effect becomes less significant over time after the initial relationship is terminated. In principle, learning mechanisms could be different for floods and heat exposure. Floods occur less frequently but are more destructive and salient. While our results thus far are consistent with similar responses, it seems important to better understand learning processes in investment decisions of firms when it comes to gradual trends and less frequent but salient shocks in the context of climate change.

6. Conclusion

This paper studies the question of whether firms adjust their supply-chain networks as a result of experienced increases in their suppliers' exposure to heat as one of the most pervasive climate-change-related hazards. To address the question, we combine granular data on global supply chains with meteorologic records of high temperatures and climate projections. We document three main insights.

First, the financial performance of suppliers in our sample is negatively affected by heat, and the consequences propagate to customers through supply chain links. Second, firms seem to terminate relationships when adverse weather at the locations of their suppliers becomes more frequent. Consistent with experience-based learning, this effect increases with signal strength and repetition and is stronger for suppliers in competitive industries and weaker for closely integrated supply-chain relationships. Third, customers choose replacement suppliers with lower expected exposure to climate hazards.

Before terminating relationships, customers may push for changes in prices and order quantities, or require additional contract provisions and insurance. The role of insurance in particular could be an important area for future

research as it could mitigate or exacerbate downstream effects. For example, [Annan and Schlenker \(2015\)](#) find that the sensitivity of crop yields to heat has remained relatively constant potentially because of frictions in the pricing of policies. In addition, the adaptation efforts of internationally diversified firms could adversely affect economic development. As developing countries in particular are expected to experience adverse change, firms in less developed countries might be more likely to lose customers to suppliers in less vulnerable locations. These effects could further economically weaken the areas most vulnerable to climate change.

Appendix

Description. Table A.4 shows additional details on the cross-sectional heterogeneity with respect to the propagation of the effects of adverse weather along the supply chain. The purpose of the tests is to study the economic mechanisms behind our findings in Section 2.2. The paper contains a condensed version of the discussion. In all tests, we aggregate the number of heat days across suppliers over the contemporaneous and previous three quarters, and interact this variable with the mean supplier, customer, and firm-pair characteristics outlined above. All cross-sectional characteristics are lagged by one year to address concerns that our explanatory variables themselves could be affected by the weather.

First, all else equal, the effect of suppliers' exposure to heat on customer firm performance could be expected to increase with the magnitude of supplier disruptions. Hence, we test if the propagation effect is larger more vulnerable suppliers. The literature has documented that heat particularly affects firms with high labor intensity ([Sepannen et al. 2006](#); [Xiang et al. 2014](#)). Consistent with this notion, we find a significantly stronger propagation effect of high temperatures for suppliers in the agricultural, mining, and construction sectors in column 2. Further, we do not find a significant interaction effect of heat days and supplier asset tangibility.

Second, we expect to find a pronounced propagation effect of heat if suppliers are able to (partially) pass on the related costs downstream, or if customers are unable to mitigate supply-chain disruptions. To test this idea, we construct the following measures of mitigating measures and relative power in supply chain relationships: 'supplier industry competitiveness', that is, the number of firms in the supplier's SIC two-digit industry from *Worldscope*, captures relative supplier bargaining power, 'industry-level input concentration', that is, the HHI across input industries from the BEA input-output matrices for a given customer industry, 'customer inventory', that is, the scaled customer inventory from *Worldscope*, and 'supplier diversification', that is, the number of suppliers for a given customer scaled by the number of unique supplier SIC two-digit industries. 'Sales correlation', that is, the moving correlation in supplier and customer sales over the past 9 quarters, and 'relationship length', that is, the number of years since the initiation of the supply-chain relationship, capture the depth of integration between a supplier and customer.

Focusing on input substitutability and customer dependence, we find a statistically significant moderating effect of supplier industry competitiveness on the propagation of heat in column 3. This finding indicates that customers' ability to switch suppliers and low supplier bargaining power mitigates customer exposure to the effects of heat in their supply chains. Similarly, we find that shock propagation is exacerbated when the customer industry more heavily relies on inputs from a single supplier industry (column 4), and mitigated for high customer inventory holdings (column 5), and supplier diversification (column 6), albeit not statistically significant. Further, we do not find a significant effect of customers' and suppliers' sales correlation but a positive effect for long relationship length (column 8). These effects are consistent with the idea that customers with a stronger reliance on their suppliers' inputs are more exposed to supplier weather hazards, and that customers and suppliers in a more closely integrated relationship (that is, longer relationship length) are able to better mitigate the repercussions.

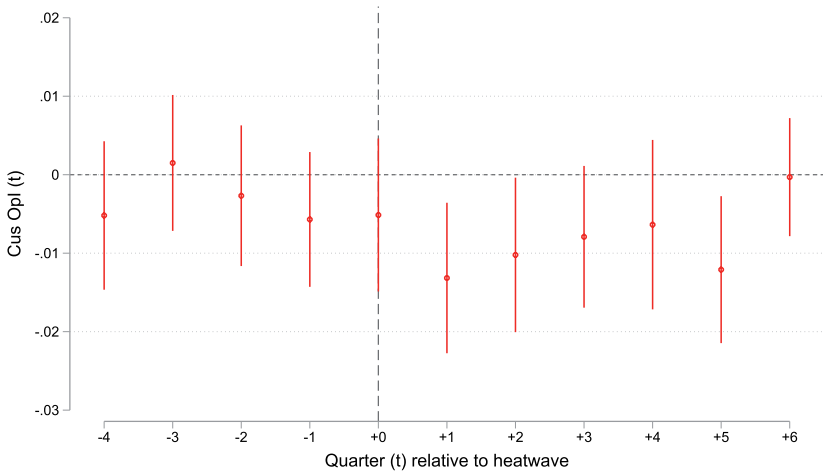


Figure A.1
Customer operating income around supplier heatwaves

This figure shows the dynamic effect of heat at the supplier firm locations on customer operating income. The dependent variable, customer operating income, is scaled by one-year-lagged assets and multiplied by 100. Specifically, the plot shows the coefficient estimates and corresponding 95% confidence intervals for β_t from Equation (2) with $t \in [-4; 6]$ indicating the quarterly time period relative to the occurrence of a heatwave. To facilitate the interpretation of the dynamic effect of the occurrence of heat relative to customer operating income, we use indicator variables of heatwaves instead of the continuous exposure (as in our main specification) over the fiscal quarters in this figure. Each model includes customer-by-quarter fixed effects, industry-by-year-by-quarter (two-digit SIC) fixed effects, country-specific time-trends, and size, age, and profitability by time fixed effects as in Barrot and Sauvagnat (2016), and is estimated on the full length of the supplier-customer relationship.

Table A.1
Variable definitions and data sources

Variable	Description
1 (Last relationship year)	Indicator variable that takes the value of one if the current year is the last active year of the supplier-customer relationship. The last year in the data is removed to avoid mechanical relationship endings (<i>Source</i> : FactSet Revere)
1 (<i>Realized</i> > <i>Expected</i>) (Heat / Flood days)	Indicator variable that takes the value of one if the number of heat days or flood days in a given year exceeds the expected (i.e., average) number of heat or flood days from a benchmark period before the initiation of the supplier-customer relationship. Details on the variable construction are provided in Section 3.1 and Figure 4. (> 1) in the variable name indicates that we drop the first year of the relationship (<i>Sources</i> : FactSet Revere, European Center for Medium-term Weather Forecasts [ECMWF], Dartmouth Flood Observatory [DFO])
Heat days	The number of days in a given firm-quarter during which the local daily high temperature exceeded the threshold of 30° C (<i>Source</i> : ECMWF)
Flood days	The number of days in a given firm-quarter during which the location was inundated according to flood maps (<i>Source</i> : DFO)
Pct. sales sup (%)	The percentage of sales represented by the customer firm in our sample relative to total sales of the supplier firm (<i>Sources</i> : FactSet Revere)
Pct. COGS cus (%)	The percentage of COGS of the customer firm represented by the sales from the supplier to the customer in our sample (<i>Source</i> : FactSet Revere)
Sup-cus HQ distance	The distance between the supplier firm's and customer firm's headquarter in kilometers (<i>Sources</i> : FactSet, Worldscope)
Total assets	Book value of assets (AT) in billions of USD (<i>Source</i> : Worldscope)
Market cap	Market Capitalization, that is, $\text{Share price} \times \text{\#shares outstanding}$, in billions of USD (<i>Source</i> : WorldScope)
ROE	Net Income scaled by average of last year's and current year's common equity (<i>Source</i> : Worldscope)
Op. income / assets	Operating income scaled by lagged total book assets and multiplied by 100 (<i>Source</i> : Worldscope)
Revenue / assets	Revenue scaled by lagged total book assets and multiplied by 100 (<i>Source</i> : Worldscope)
Asset tangibility	Property, Plants, and Equipment (PPE) scaled by total book assets (<i>Source</i> : Worldscope)
Ind. vulnerability	Indicator variable that takes the value of one if the firm is operating in an industry vulnerable to climate-related shocks, that is, agriculture, mining, and construction (SIC two-digit codes 1–9, 10–14, 15–17) (<i>Source</i> : Worldscope)
Same SIC2 industry	Indicator variable that takes the value of one if both customer and supplier are operating in the same SIC two-digit industry (<i>Sources</i> : FactSet Revere, Worldscope)
Same country	Indicator variable that takes the value of one if customer and supplier are headquartered in the same country (<i>Source</i> : FactSet Revere, Worldscope)
Relationship length	The number of years between the current year of the given observation and the year in which the customer-supplier relationship was first established (<i>Source</i> : FactSet Revere)
No. of customers	The number of customers for a given supplier firm in a given year (<i>Source</i> : FactSet Revere)
No. of suppliers	The number of suppliers for a given customer firm in a given year (<i>Source</i> : FactSet Revere)
No. of suppliers / assets	The number of suppliers for a given customer firm in a given year, scaled by the firm's lagged total book assets (<i>Source</i> : FactSet Revere)
Supplier diversification	The number of suppliers for a given customer firm in a given year, scaled by the number of unique SIC two-digit industries across all of the firm's suppliers (<i>Source</i> : FactSet Revere)
N firms ind.	Industry-competitiveness, measured as the number of public firms globally in the same SIC two-digit industry code, in thousands (<i>Source</i> : Worldscope)
HHI sales ind.	Input-Industry concentration, measured as the Herfindahl-Hirschman Index (HHI) of revenues in the firm's SIC two-digit industry (<i>Source</i> : Worldscope)
Sup-to-cus ind. sales	Value of sales from the supplier industry to the customer industry (in USD) (<i>Source</i> : Bureau of Economic Analysis [BEA])

(Continued)

Table A.1
Continued

Variable	Description
HHI ind. inputs	Herfindahl-Hirschman Index (HHI) across values of inputs provided by supplier-industries for a given customer-industry from the BEA Input-Output matrices. BEA Input-Output matrices are provided at the NAICS5 industry level (<i>Source</i> : BEA)
Sales correlation	The rolling correlation between sales of the customer and supplier firm at the quarterly frequency over the last nine quarters (<i>Sources</i> : Worldscope, Datastream)
Acct. receivable / assets	Accounts Receivable, scaled by the firm's lagged total book assets (<i>Source</i> : Worldscope)
R&D / assets	Research and Development (R&D) expenses, scaled by the firm's lagged total book assets (<i>Source</i> : Worldscope)
COGS / assets	Cost of Goods Sold (COGS), scaled by the firm's lagged total book assets (<i>Source</i> : Worldscope)
Inventory / assets	Total inventory, scaled by the firm's lagged total book assets (<i>Source</i> : Worldscope)
Cash / assets	Cash holdings, scaled by the firm's lagged total book assets (<i>Source</i> : Worldscope)
Same SIC2 industry	Indicator variable that takes the value of one if both customer and supplier are operating in the same SIC two-digit industry (<i>Source</i> : FactSet Revere, Worldscope)
1(Delisted)	Indicator variable that takes the value of one if the firm was delisted in the following fiscal year, and zero otherwise (<i>Source</i> : Worldscope)
1(AT-10%)	Indicator variable that takes the value of one if the firm's book assets (AT) declined by at least 10% in the following year, and zero otherwise (<i>Source</i> : Worldscope)
1(PPE-10%)	Indicator variable that takes the value of one if the firm's property, plants, and equipment (PPE) (i.e., tangible assets) declined by at least 10% in the following year, and zero otherwise (<i>Source</i> : Worldscope)
1(S&P / Moodys ↓)	Indicator variable that takes the value of one if the was downgraded by S&P or Moodys in the following year, and zero otherwise (<i>Source</i> : SDC Platinum)
1(Disrupted)	Indicator variable that takes the value of one, if the supply-chain relationship was interrupted in the following but later resumed, and zero otherwise (<i>Source</i> : FactSet Revere)
Sup. SIC2 (sup. SIC3)	For a given supplier-customer relationship, this measure counts the number of other suppliers in the same SIC two-digit (three-digit) industry as the focal supplier who are currently in a supplier-customer relationship with the focal customer
Acq. SIC2 (acq. SIC3)	For a given supplier-customer relationship, this measure counts the number of acquisitions announced by the focal customer in the same SIC two-digit (three-digit) industry as the focal supplier
Adaptation readiness	Indexes capturing the readiness for climate change adapatation at the country-level with respect to Economic, Social, and Governance aspects. Overall climate change adaptation readiness is a weighted average of the three aspects (<i>Source</i> : Notre-Dame Global Adaptation [ND-GAIN] index)

Table A.2
Alternative measures of heat exposure

(a) Supplier operating performance			(b) Customer operating performance		
	(1)	(2)	(1)	(2)	(3)
Heat Days (30-95) (t,t-3)	-0.0023** (-2.16)		-0.0005*** (-3.198)		
Heatwave (30/7) (t,t-3)		-0.0502* (-1.95)			
Heatwave (30-95/7) (t,t-3)				-0.0109*** (-4.141)	
Firm × Fiscal-Qtr FE	Yes	Yes	Yes	Yes	-0.0213*** (-3.509)
Ind × Year-Qtr FE	Yes	Yes	Yes	Yes	Yes
Ctry-Linear-Trends	Yes	Yes	Yes	Yes	Yes
BS2016 FE	Yes	Yes	Yes	Yes	Yes
Observations	202,438	202,438	123,700	123,700	123,700
Customers	5,628	5,628	6,299	6,299	6,299
R ²	.631	.631	.711	.711	.711

This table shows robustness tests analogous to Table 2 on the impact of heat exposure at the location of the supplier firms on their operating income (Opt) (Panel A.2a) and the operating income of their downstream customers (Panel A.2b). The dependent variable is scaled by one-year-lagged assets and multiplied by 100. The alternative measures of heat exposure are *Heat Days* (30–95), that is, the number of days with temperatures above 30° C and in the 95th percentile of historical local temperatures, *Heatw* (30/7), that is, a dummy for the occurrence of 7 consecutive days with temperatures above 30° C, *Heatw* (30–95/7), that is, a dummy for the occurrence of >= 7 consecutive days with temperatures above 30° C and in the 95th percentile of historical local temperatures. All three measures are aggregated over the current and the three preceding financial quarters (*t*–3 to *t*). The number of observations refers to supplier firm-quarters in Panel A.2a and to customer firm-quarters in Panel A.2b. The sample period is 2003 to 2016. We exclude firms in the financial industry as well as firms with less than 10% of firm locations within 30 km of the headquarters. All regressions include firm-by-fiscal quarter fixed effects, industry-by-year-by-quarter fixed effects, industry-specific time fixed effects, and country-specific linear trends. We additionally include interaction terms of terciles of firm size, age, and ROA with year-by-quarter fixed effects (i.e., BS2016 FE), following Barrot and Sauvagnat (2016) as indicated. Standard errors are clustered on the firm level. **p*>.1, ***p*>.05, ****p*>.01.

Table A.3
Location-specific exposure and downstream propagation

(a) Across locations			(b) Placebo test		
	Cus Opl (t)			Cus Opl (t)	
	(1)	(2)		(1)	(2)
Sup Heat Days (30) (t-0) (z)	-0.0142 (-0.611)	-0.0006 (-0.027)	Sup Heat Days (30) (t-0) (z)	-0.0033 (-0.120)	-0.0225 (-0.821)
Sup Heat Days (30) (t-1) (z)	-0.0494** (-2.205)	-0.0472** (-2.112)	Sup Heat Days (30) (t-1) (z)	0.0294 (1.013)	0.0365 (1.223)
Sup Heat Days (30) (t-2) (z)	-0.0128 (-0.671)	-0.0104 (-0.540)	Sup Heat Days (30) (t-2) (z)	0.0454* (1.935)	0.0338 (1.358)
Sup Heat Days (30) (t-3) (z)	-0.0301* (-1.840)	-0.0306* (-1.842)	Sup Heat Days (30) (t-3) (z)	0.0447* (1.828)	0.0331 (1.279)
Firm × Fiscal-Qtr FE	Yes	Yes	Firm × Fiscal-Qtr FE	Yes	Yes
Ind × Year-Qtr FE	Yes	Yes	Ind × Year-Qtr FE	Yes	Yes
Ctry-Linear-Trends	Yes	Yes	Ctry-Linear-Trends	Yes	Yes
BS2016 FE	No	Yes	BS2016 FE	No	Yes
Observations	85,575	85,575	Observations	169,862	169,854
Customers	5,026	5,026	Firm-Pairs	5,591	5,591
R ²	.715	.721	R ²	.588	.593

This table presents OLS regression estimates on the effects of heat days across locations of suppliers on their customers. The dependent variable in both panels is customer Operating Income (Opl), scaled by one-year-lagged assets and multiplied by 100. *Sup heat days* (30) (*t*)(*z*) is a continuous variable that sums over the total number of heat days across all supplier locations in a given quarter. Because of the substantial variation in the numbers of observed locations per firm, we standardize our main variables of interest to have mean of zero and standard deviation of one in this test, indicated by (*z*) in the variable label. In Panel (a), relationships between customers and suppliers are active. Panel (b) implements a placebo test by repeating the analysis on a sample of supplier-customer years in which the relationships were inactive according to FactSet Revere. The unit of observation is at the customer-by-year-quarter level and the sample period is from 2003 to 2016. All regressions include customer firm-by-quarter fixed effects as well as industry-by-year-quarter fixed effects. Column (2) additionally includes terciles of size, age, and ROA interacted with year-by-quarter fixed effects (BS2016 FE). *t* values in parentheses are calculated from standard errors clustered at the customer-firm level. **p* < .1, ***p* < 0.05, ****p* < 0.01.

Table A.4
Downstream propagation, cross-section

	Cus Opl (t)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Heat days (t:t-3)	-0.0085 (-0.73)	-0.0026 (-0.47)	-0.0450*** (-5.36)	0.0017 (0.24)	-0.0294*** (-5.20)	-0.0194* (-1.78)	-0.0204*** (-4.16)	-0.0343*** (-5.76)
Heat days (t:t-3) × Sup Tangibility	-0.0004 (-1.20)							
Heat days (t:t-3) × Sup-Ind Vuln.		-0.0012*** (-4.50)						
Heat days (t:t-3) × Sup-Ind Comp.			0.0216*** (3.05)					
Heat days (t:t-3) × Input-Ind Conc.				-0.0588*** (-2.93)				
Heat days (t:t-3) × Cus Inventory					0.0013*** (2.78)			
Heat days (t:t-3) × Sup Divers.						0.0013 (0.28)		
Heat days (t:t-3) × Sales Corr.							-0.0000 (-0.18)	
Heat days (t:t-3) × Rel. Length								0.0054*** (3.66)
Sup Tangibility	0.1260 (0.93)							
Sup-Ind Vuln.		0.2382** (2.06)						
Sup-Ind Comp.			-8.1227* (-1.85)					
Cus Inventory					1.1015* (1.84)			
Sup Divers.						-11.4184*** (-2.97)		
Sales Corr.							-0.0847* (-1.79)	
Rel. Length								0.3625 (0.27)
Firm × Fiscal-qr FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind × Year-qr FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ctry-linear-trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
BS2016 FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	117,039	123,700	102,760	54,187	114,095	122,162	93,022	123,700
R ²	.709	.711	.713	.717	.708	.710	.712	.711

This table presents OLS regression estimates on cross-sectional differences in the effects of heat days at the location of the suppliers on customer performance. The dependent variable in both panels is customer Operating Income (Opl), scaled by one-year-lagged assets and multiplied by 100. *Heat days* ($t, t-3$) measures the total number of heat days at all suppliers of a given customer during the contemporaneous and previous three quarters. ‘Sup Tangibility’ is the ratio of assets to plants, property and equipment, ‘Supplier Industry Vulnerability’ is a dummy that takes the value of one for climate-exposed industries, ‘Supplier-Industry Competition’ is the number of firms in the supplier industry, ‘Input-Industry Concentration’ is the HHI of input industries according to the BEA input-output matrices for a given customer industry, ‘Customer Inventory’ is the scaled value of customer firm inventory, ‘Supplier Diversification’ is the number of supplier firms scaled by the number of unique supplier industries for each customer, ‘Sales Correlation’ is the rolling correlation in sales between customer and supplier over the last 9 quarters, ‘Relationship Length’ is the number of years since the start of the supplier-customer relationship. The data are organized at the customer-year-quarter level and the sample period is 2003 to 2016. The specification and filters follow Table 2. All regressions include fixed effects as indicated. * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Table A.5
Downstream propagation: Other outcomes

	(1) Rev/Assets (t)	(2) Rev/Empl (t)	(3) RevGrowth (t)	(4) OpMargin (t)	(5) AccPayable (t)	(6) COGS (t)	(7) Inventory (t)
Heat days (t-0)	-0.0011*** (-3.23)	-0.0082* (-1.68)	-0.0000 (-1.01)	-0.0005 (-1.21)	-0.0004** (-2.04)	-0.0006** (-2.20)	-0.0002 (-1.22)
Heat days (t-1)	-0.0018*** (-4.18)	-0.0193*** (-3.40)	-0.0000*** (-3.93)	-0.0011** (-2.04)	-0.0005** (-2.22)	-0.0014*** (-3.38)	-0.0004*** (-2.63)
Heat days (t-2)	-0.0011*** (-2.95)	-0.0023 (-0.52)	-0.0000 (-0.40)	-0.0003 (-0.47)	-0.0001 (-0.69)	-0.0007** (-2.29)	-0.0002 (-1.15)
Heat days (t-3)	-0.0012*** (-3.19)	-0.0069 (-1.50)	-0.0000*** (-3.54)	-0.0007* (-1.80)	-0.0000 (-0.23)	-0.0007** (-2.46)	-0.0003** (-2.14)
Firm × Fiscal-qr FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind × Year-qr FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ctry-linear-trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes
BS2016 FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	123,699	83,646	96,179	122,946	101,948	112,581	111,636
Customers	6,299	4,257	4,823	6,227	5,635	5,835	5,851
R ²	.886	.899	.307	.745	.886	.913	.936

This table presents OLS estimates on the effect of heat at the location of the suppliers on several customer firm-level outcomes. The dependent variables are customer revenues over asset (column 1), revenues over number of employees (column 2), revenue growth (column 3), operating margin (column 4), accounts payable (column 5), cost of goods sold (column 6), and inventory (column 7). The accounts payable, COGS, and inventory are scaled by one-year-lagged total assets. The data are organized at the customer-year-quarter level. The independent variables capturing the number of heat days across the suppliers of a given customer in quarters t to $t-3$. The specification and filters follow Table 2. Standard errors are clustered at the firm level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6
Heat and natural disasters: Direct and indirect effects

(a) Direct effects of heat days on suppliers					
	(1)	(2)	(3)	(4)	(5)
Heat Days (t,t-3)	-0.00257*** (-2.71)				
ex. Heatwave (t,t-3)		-0.00284*** (-3.33)			
ex. Fire (t,t-3)			-0.00250*** (-3.17)		
ex. Drought (t,t-3)				-0.00164** (-2.06)	
ex. Heatwave/Drought/ Fire (t,t-3)					-0.00216*** (-3.26)
Firm × Fiscal-qr FE	Yes	Yes	Yes	Yes	Yes
Ind × Year-qr FE	Yes	Yes	Yes	Yes	Yes
Cry-linear-trends	Yes	Yes	Yes	Yes	Yes
Observations	202,438	202,438	202,438	202,438	202,438
Customers	5,628	5,628	5,628	5,628	5,628
R ²	.626	.626	.626	.626	.626

(b) Indirect effects of heat days on customers					
	(1)	(2)	(3)	(4)	(5)
Heat Days (t,t-3)	-0.00023*** (-5.15)				
ex. Heatwave (t,t-3)		-0.00019*** (-4.54)			
ex. Fire (t,t-3)			-0.00022*** (-5.48)		
ex. Drought (t,t-3)				-0.00021*** (-4.99)	
ex. Heatwave/Drought/ Fire (t,t-3)					-0.00021*** (-4.99)
Firm × Fiscal-qr FE	Yes	Yes	Yes	Yes	Yes
Ind × Year-qr FE	Yes	Yes	Yes	Yes	Yes
Cry-linear-trends	Yes	Yes	Yes	Yes	Yes
Observations	123,700	123,700	123,700	123,700	123,700
Customers	6,299	6,299	6,299	6,299	6,299
R ²	.707	.707	.707	.707	.707

This table presents OLS regression estimates on the effects of heat at the location of supplier firms on customer operating performance, accounting for the contemporaneous occurrence of natural disasters recorded by EM-DAT. Table A.6a shows the direct effects on suppliers' operating income (Opl) scaled by lagged assets and multiplied by 100 for ease of interpretation. Table A.6b shows the corresponding results for customers. *Heat Days (t)* indicates the total number of heat days at the supplier firm in column (1), and after deducting the number of days on which the home country of the supplier was affected by heatwaves (column 2), wildfires (column 3), droughts (column 4), or any of these events (column 5). We estimate the effect of the sum of days during the financial quarter *t* and the three preceding quarters (*t* - 3 to *t* - 1), with the number of observations referring to supplier firm year-quarters in Panel A.6a and customer firm year-quarters in Panel A.6b. Consistent with the main tests, we estimate regressions as outlined in Equation (1). * $p < .1$, ** $p < .05$, *** $p < .001$.

Table A.7
Alternative to ordinary least squares

	Cox Hazard - 1 (Last rel. year)	
	(1)	(2)
$\mathbb{1}(\text{Real.} > \text{Exp. heat days})$	-0.041** (-2.239)	-0.052*** (-2.699)
Sup-ind.-year strata	Yes	Yes
Cus-ind.-year strata	Yes	Yes
Sup-ctry-cus-ctry-year strata	No	Yes
SE cluster	Rel.	Rel.
N	114,321	114,321

This table shows Cox proportional hazard model regression estimates (not hazard ratios) on the effects of realized versus expected weather on supply-chain relationship termination. The dependent variable is an indicator variable that takes the value of one if the supply-chain relationship ended after the current year. The start of a supplier-customer relationship is the first year the relationship is documented in the FactSet Revere database, the end is the year a relationship is terminated. We drop relationships that were terminated and subsequently restarted at some point in our sample. Following [Fee et al. \(2006\)](#), if a relationship lasts until the final year of the sample period, we treat the duration of relationship as being right-censored. The main independent variable $\mathbb{1}(\text{Realized} > \text{Expected})$ (> 1) is an indicator variable that takes the value of one in year t if the difference between the realized number of heat days per year since the beginning of the supply-chain relationship exceeds the expected number of days, and zero otherwise. The specification and data filters follow [Table 3](#). The unit of observation is at the relationship-year level. Robust standard errors are clustered at the relationship level. * $p < .1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8
Alternative estimation periods

(a) Alternative expected exposure estimates, 5 years before relationship			
	Dep. var.: $I(\text{Last rel. year})$		
	(1)	(2)	(3)
$I(\text{Real.} > \text{Exp. heat days}) (> 1)$	0.0813*** (7.258)	0.0812*** (7.070)	0.0770*** (7.201)
Sup.-ctry-year FE	Yes	Yes	Yes
Sup.-ind.-year FE	No	Yes	Yes
Cus.-ind.-year FE	No	Yes	Yes
Sup-ctry-cus-ctry-year FE	No	No	Yes
Observations	116,412	116,412	116,412
R^2	.0752	.0834	.1358
(b) Alternative expected exposure estimates, 15 years before relationship			
	Dep. var.: $I(\text{Last rel. year})$		
	(1)	(2)	(3)
$I(\text{Real.} > \text{Exp. heat days}) (> 1)$	0.0869*** (7.329)	0.0875*** (7.317)	0.0829*** (7.191)
Sup.-ctry-year FE	Yes	Yes	Yes
Sup.-ind.-year FE	No	Yes	Yes
Cus.-ind.-year FE	No	Yes	Yes
Sup-ctry-cus-ctry-year FE	No	No	Yes
Observations	116,412	116,412	116,412
R^2	.0766	.0849	.1370

Analogous to Table 3, this table presents linear probability model estimates on the impact of the exceedance of heat exposure expectations on the likelihood of supply-chain relationship termination. The sample and variables are constructed as in Table 3. The main difference to Table 3 is that we use benchmark periods of 5 and 15 years before the establishment of a supply-chain relationship to construct our main variables of interest, $I(\text{Realized} > \text{Expected})(> 1)$. The specification and filters follow in Table 3. The regressions include relationship fixed effects, year fixed effects, supplier and customer-industry-by-year, as well as supplier-country-by-customer-country-by-year fixed effects as indicated. Robust standard errors are double-clustered at the relationship and year level. * $p < .1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9
Anecdotal sample of supply agreements from SEC EDGAR

CIK	Start date	Term (years)	Automatic renewal (0/1)	Renewal (months)	Convenience clause (0/1)	Termination notice (Days)	Force Majeure clause (0/1)
46216	Dec. 10, 2004	5.00	No	12	Yes	90	Yes
789132	Jan. 27, 2015	2.00	No	3	Yes	30	Yes
808362	Nov. 13, 2018	5.00	Yes				No
867773	Feb. 14, 2022	1.00	No		No		Yes
884713	Jan. 1, 2011	6.00			No		Yes
906709	Oct. 29, 2010	10.00			Yes		Yes
1002811	May. 24, 2011	10.00	Yes		No	90	Yes
1023362	Nov. 1, 2002	2.00	Yes	6	Yes	180	No
1054102	Aug. 13, 2014	0.00	Yes	12	No		Yes
1056358	Jun. 1, 2008	3.00	Yes	3	No		Yes
1063697	May. 1, 2008	2.00	No		No	60	No
1092050	Feb. 23, 2005	0.50	No	1			No
1100962	Apr. 27, 2012	4.00	Yes				Yes
1158091	Nov. 18, 2005	indefinite	No		Yes	60	Yes
1272547	Apr. 4, 2004	2.00	Yes	2	No	30	Yes
1318605	Sep. 1, 2006	3.00	Yes	6	Yes	90	Yes
1371541	Apr. 11, 2015	0.00			Yes	5	Yes
1407038	Mar. 20, 2009	1.00	Yes	6	Yes	90	Yes
1455684	Sep. 6, 2007	8.00	No		Yes	30	Yes
1455684	Oct. 1, 2013	3.00	No		Yes	30	Yes
1455684	Sep. 28, 2016	4.00			Yes	30	Yes
1516479	Mar. 7, 2011	5.00	No		No		Yes
1544229	Jul. 1, 2012	indefinite			Yes		Yes
1546640	redacted	2.00	Yes	2	Yes	180	Yes
1588972	Jan. 1, 2019	5.00	Yes		Yes	720	Yes
1679082	May. 30, 2018	5.00	Yes	12	Yes	180	Yes
1808997	Aug. 24, 2020	2.00	No	6	No		Yes
Mean		3.62	0.55	5.92	0.63	118.44	0.85
SD		2.76	0.51	4.08	0.49	170.35	0.36

This table shows anecdotal information on contract details and provisions of 27 individual supply-chain contracts extracted from corporate disclosure on supplier-contracts in 10-K forms. The contract details and provisions were obtained by manually searching the SEC EDGAR platform for the term “supply agreement.” ‘CIK’ is the EDGAR identifier of the firm that disclosed the supplier-contract to the SEC. ‘Start Date’ is the beginning date of the contract. ‘Term’ (in years) is the length of the supplier contract. ‘Automatic Renewal’ indicates whether the contract will automatically renew at the frequency indicated in column ‘Renewal (months)’ unless it is explicitly terminated. ‘Convenience Clause’ indicates that the customer has the right to terminate the supplier with a notice time indicated in column ‘Termination Notice (Days)’. ‘Force Majeure Clause’ indicates that the customer has the right to terminate the supplier in case of a force majeure event, such as a global pandemic, war in the supplier country, or the occurrence of a natural disaster. Blank fields were redacted, were unavailable, or do not apply for the given contract.

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